Nm Index: A Percentile Score by Considering Author Contribution, Ordering, Influential Self Citation, Career Stage, and Domain Norms for Global Individual Scholarly Impact Evaluation

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Abstract

Although some scores like C-scores consider author position, they overlook actual contributions, especially for middle authors or when ordering norms don't reflect true input. Moreover, position-based scores often include redundancies and over-rewards, such as repeating the single-author factor across three sub-scores. Citations also vary by domain and publication year, yet the C-score ignores outlier effects—like a few papers with extremely high citations or many with low ones—and multi-domain involvement. Similarly, H, g, Hf, and Hm indices fluctuate based on paper domains and career stage, disadvantaging low-citation fields or early-career researchers. Finally, the m-index overcompensates for career adjustment, penalizing seniors, while Hm and Hf assume equal author contributions. To address these shortcomings, this paper proposes a novel mathematical model for Nilmantha's index (Nm-index), a percentile score indicating scholarly impact directly comparable globally. It accounts for author ordering norms and contributions via variables tracking: Alphabetical/Random (A/R), Relative Contribution (RC) with multiple/single First Authors (FAs) or Last Authors (LAs), and Explicit Contribution (EC) with multiple/single FAs/LAs. Specifically, it uses uniform weights for A/R, proportional uniform weights for multiple FAs/LAs per RC/EC, a harmonic model for RC, and AHP-driven CRediT or ACI. First, citations are adjusted for influential self-citations, considering self-influence and overlapping core-author influence. Second, we compute outlier-uncontrolled Weighted field- and Year-Normalized (WFYN) total cites (overall impact), outlier-controlled WFYN citations per paper (average paper impact), outlier-controlled WFYN citation rate (impact rate), and weighted average hf (productive cited output with fractional credit), hm (productive cited output for multi-authorship), and g (highly cited works) indices—all compensated for career factor (preserving senior academics' collective achievements) and multiple fields. The model is validated using fictitious author profile use cases. For numerical validation, we analyze statistics of each proposed factor, C-score, H-index, and Cm-score across 100,000 Google Scholar profiles in 334 fields—for the whole sample, disciplines, and career stages. Finally, we evaluate paired differences between metrics, intra-class correlation for interchangeability, discriminative power for award winners, and Nm-index's unique predictive power via multiple linear regression on influential self-citations, author contributions and norms, domain sensitivity, career stage, and temporal evolution.

Keywords: Author contribution, ordering, domain norms, self-citation, career compensation, Nm-index

1 Introduction

1.1 Background

In academic field, research evaluation is important because it affects choices about things for the academics like faculty promotions, funding distribution, and institutional rankings [1]. Therefore, bibliometrics has actually become essential in order to measure research impact by using citation-based indices as scholarly output, which has increased with the digital age [2]. However, we see that the diversity among disciplines, the rise in multi-author papers, and the differences in career stages can present as serious obstacles to these kind of metrics and thus can result in unfair or unjustifiable comparisons among scholars [3]. So, we can point out that the need for trustworthy metrics in order to manage this enormous output has increased during recent times due to the exponential growth in scientific publications, which is expected to double every 17 years at an annual rate of about 4-5% according to research [4]. Moreover, we can observe that in an increasingly competitive environment, this proliferation highlights the need for metrics that not only measure impact but also can adjust to changing academic practices in order to ensure fair assessments [5].

So as to weighing productivity and impact, the Hirsch index, which is also known as the H-index, became a fundamental metric for evaluating individual researchers over the past two decades. Specifically, note that it denotes that a given scholar has an H-index of h, if and only if h of their papers have at least h citations each [1]. Because of its simplicity and easiness of using in systems, it has been widely used in scholarly impact evaluations [6]. Moreover, we can denote that its integration into platforms such as Google Scholar and Scopus, which enable stakeholders to conduct rapid evaluations, indeed has demonstrated its widespread adoption in the academic field [7]. On the other hand, we can highlight that recent criticisms, however, point out that the time-dependency of the H-index can inflate scores for researchers with long careers without taking current productivity into account [8]. In addition, it can be pointed out that other noteworthy drawbacks of the H-index include its insensitivity to extremely highly cited works, biases toward high-citation fields, disregard for author positions, equal treatment of co-authors regardless of contribution, and disadvantages for early-career researchers [3, 9].

It should be noted that a number of H-index extensions have been suggested by several experts in the field in an effort to address these problems. First, let's consider the g-index, which highlights highly cited papers [10] and it is defined as the largest number g such that the top g papers, which are written by the author receive at least g^2 citations cumulatively. Next, by employing a harmonic mean in the process of computing effective paper rank so as to fractionally distribute citations according to author count for each paper, which is written by the author, the Hm-index accounts for co-authorship [11]. Similarly, so as to more fairly handle multi-authorship, another variant of H-index, which is called as the Hf-index uses fractional allocation [12]. Next, in order to account for career length, we see that the m-index, which is also known as the m-quotient, actually simply divides the H-index by the number of years starting from the first publication year [1]. Although we can understand that these variations address certain shortcomings, such as some considerations for collaboration (Hm, Hf) or rewarding peak impacts (g-index), they frequently maintain the presumption of equal contributions, over- or under-compensate for career stages, and continue to be sensitive to disciplinary citation norms [2, 13]. Although they try to address temporal biases, other variations, like the AR-index, which takes citation age into account, still have trouble with interdisciplinary comparisons [14]. Some have also recently proposed and index called GK index that uses data analysis to identify hidden patterns in a dataset using machine learning techniques and disjointness [15]. Despite these efforts, empirical studies reveal that many variants have low discrimination power, frequently failing to differentiate between researchers with similar profiles [16].

Current self-citation calculation practices in bibliometrics primarily focus on author self-citations, where an individual cites their own work, and co-author self-citations, which account for citations from any overlapping authors [17]. The most widely used metric, the author self-citation rate, measures the proportion of an author's citations that reference their own publications, offering a clear indicator of referencing behavior [18]. Scopus employs author IDs to compute self-citations, providing options to exclude them, while Web of Science uses name matching, often reporting citation counts without self-citations for refined analysis [19, 20]. Another index called Fi-score attempts to evaluate the self-citation effect by using the H-index and total citations and by comparing with a data driven-threshold [21]. Similar to that, some indices called the b-index [22] accounts s-index for self-citations. So, as to conclude, we find that these reviewed methods most importantly do not properly track the

actual influence for the self citation, which we consider the key parameter that should be considered in discounting the proportion of the citation. For instance, in the current author self-citation, the author will receive 100% self citation, even if that author in the citing paper may have a very low contribution with many non-overlapping authors present in the citing paper, which implies that a low probability that the author himself may have cited his own work (it is more likely cited by the non-overlapping author). On the other hand, in the core-author self citation also, the author receives 100% self citation even if overlapping core authors cite the work despite there being a low chance of the author may have influenced the core authors to cite his work. We point out that 100% penalty is applicable when there is actually 100% overlap of authors between the citing and the cited paper. Thus, we attempt to capture an approximate **influence**, which should be the factor accounted in discounting the self-citation.

We note that more sophisticated metrics, just like Ioannidis' composite C-score infact combine several indicators. Specifically, it combines total citations, H-index, Hm-index, and authorship positions (single, first, and last) as in the form of roles rather than the true contribution, to offer some advanced evaluation of scholarly impact [23]. It should be noted that this strategy has indeed outperformed individual conventional metrics like the H-index and has been considered as a breakthrough for prioritizing key author roles [24]. In fact, this has been adapted by major scientific databases like Scopus at the time of this writing. The C-score, however, we can point out that it lacks adaptability to different authorship norms, such as alphabetical ordering, and demonstrates redundancies, such as over-rewarding single authorship across sub-scores. In addition, note that it also fails to capture the true contributions of intermediate authors [25]. Furthermore, we see that it ignores biases that are resulting from outlier citations, domain-specific citation patterns, and inadequate career-stage adjustments, which may disadvantage junior or interdisciplinary scholars [13, 26]. We also can further point out that the composite nature of the C-score has also been questioned by recent analyses, which contend that just by adding indicators without psychometric validation might not produce a useful unified measure [27]. Finally, its dependence on Scopus data may introduce biases in coverage, which is especially true for emerging fields or non-English publications [28].

1.2 Problem Statement

The proliferation of bibliometric indices has paradoxically created a crisis of scholarly assessment credibility. When academics with genuinely equivalent research impact receive dramatically different metric scores simply because one works in a high-citation field, or because one is an early-career researcher, or because disciplinary norms place them in middle author positions despite substantial contributions, the fundamental purpose of scholarly evaluation is undermined [3, 26]. This systematic mismeasurement manifests across academia: funding bodies allocate resources based on indices that conflate citation density with research quality; hiring committees compare candidates using metrics that reward gaming behaviors over genuine scholarship; and institutional rankings aggregate individual scores that fail to capture actual intellectual contributions [2, 25]. The result is a scholarly ecosystem where strategic positioning and disciplinary choice increasingly matter more than research substance, eroding trust in academic evaluation and creating perverse incentives that distort research practices and collaboration patterns [17, 29].

1.3 Research Questions

To address the identified gaps, this study investigates the following research questions:

1. **RQ1:** How can author contribution be accurately quantified across different ordering norms (alphabetical, relative, and explicit contribution) while fairly crediting intermediate authors?
2. **RQ2:** How should self-citations be discounted based on the actual influence of the author and overlapping core authors in the citing paper, rather than applying uniform penalties?
3. **RQ3:** What normalization approach can effectively control for field-specific citation densities and publication year effects while preserving recognition of exceptional achievements?
4. **RQ4:** How can career-stage compensation be balanced to support early-career researchers without diminishing the accumulated impact of senior scholars?
5. **RQ5:** Can a unified percentile-based index be constructed that demonstrates superior discriminative power, reduced sensitivity to outliers, and better equity across disciplines and career stages compared to existing metrics?

1.4 Proposed Solution

We propose Nilmantha's index (Nm-index), a novel metric that introduces improvements for author contribution norms and builds upon the existing indices, to address this. Important elements include outlier-controlled normalization based on field and publication year, a tempered career-stage factor that prevents penalizing established researchers, and variables for authorship models (e.g., alphabetical/random, relative or explicit contributions with multiple first/last authors) [30, 31]. Not only that it actually accounts for the influence involved in the self citation and discount the citations proportional to the influence, which is a better technique than existing ones. Without going into all the methodological specifics here, the Nm-index provides a preview of a more reliable, contribution-aware alternative by combining components from the H, g, Hm, Hf, and C-scores. By utilizing well-established taxonomies such as CRediT for accurate contribution mapping, this integration seeks to offer a versatile framework that can be adjusted to a variety of authorship practices [32].

1.5 Contributions

The contributions of this work are as follows:

- **Authorship norm-aware contribution weighting:** A mathematical model that reduces biases in intermediate author credits by dynamically adjusting for authorship norms (alphabetical, exact contribution-based, relative ordering, multiple leads) using corresponding harmonic/uniform/CRediT/ACI-driven weighting.
- **Outlier-controlled field and year normalization:** By integrating outlier control through field/year-specific benchmarks, logarithmic normalization, and capping, extreme citation effects and domain disparities are reduced.
- **Balanced career compensation factor:** A career adjustment metric that appropriately rewards early-stage researchers without reducing the overall influence of senior researchers.
- **Influential self-citation discounting:** Weighting of self-influence and core-author influence in the citing paper with respect to overlapping authors between the cited and citing paper, incorporating overlapping author contribution to fairly account for self-citations.
- **Six modified sub-metrics:** Modified hf-index, modified hm-index, and modified g-index adjusted for author contribution and career factor; outlier-uncontrolled field and year normalized total citations; outlier-controlled field normalized citations per paper; and outlier-controlled field and year normalized citation rate.
- **Large-scale empirical validation:** Validation using extensive datasets (100,000 profiles across 334 fields) with statistical analyses demonstrating better robustness to gaming and higher equity across disciplines and career stages compared to existing indices.
- **Comprehensive sensitivity analysis:** Empirical evaluation demonstrating reduced sensitivity to outliers, author contribution variations, field impact, career stage effects, and temporal evolution, with correlation analyses against existing assessment indices.

1.6 Novelty Statement

The fundamental novelty of this work lies in reconceptualizing bibliometric assessment as a multi-dimensional fairness problem rather than a citation aggregation exercise. While existing indices treat authorship, self-citation, field variation, and career stage as independent correction factors to be addressed separately, the Nm-index introduces a unified theoretical framework that models these as interdependent bias sources requiring simultaneous compensation. The core conceptual advance is the principle of proportional influence attribution: rather than assigning discrete categories or binary penalties, the Nm-index computes continuous influence weights that flow through all metric components, from author contribution through self-citation discounting to final percentile transformation. This proportionality principle, validated across 100,000 profiles in 334 fields, enables the first bibliometric index where a researcher's score directly reflects their actual scholarly contribution regardless of disciplinary context, career timing, or collaboration structure.

1.7 Impact and Paper Organization

The research community will benefit greatly from these innovations, which promote more fair evaluations in funding, hiring, and policy decisions [2]. The Nm-index has the potential to improve the

integrity of scholarly metrics by supporting diverse career paths and facilitating fairer interdisciplinary evaluations by better capturing genuine contributions and thwarting manipulation [26]. Our strategy may have an impact on international ranking systems and promote increased openness in authorship declarations by tackling these issues [33]. In the end, it encourages a change to responsible metrics, supporting programs such as DORA that put quality above quantity in assessments [34].

The remainder of this paper is organized as follows: Section 2 details the proposed methodology, including mathematical formulations and normalization techniques. Section 3 presents empirical validation using real-world profiles and statistical analyses. Section 4 discusses results, implications, and limitations, followed by conclusions in Section 5.

2 Material and Methods

First, let us list down the Features considered in the proposed Nm-Index.

- Uniform weight distribution among Alphabetical or Random author ordering without explicit author contribution
- Proportional uniform weight distribution in multiple First authors or Last Authors as per Relative/Explicit author contribution
- Harmonic model for relative contribution
- CRediT/ACI based weight assignment for explicit author contribution
- Outlier uncontrolled weighted field and published year normalized total cites compensated for career factor, author contribution, and multiple fields
- outlier controlled weighted field normalized citations per paper compensated for career factor, author contribution, and multiple fields
- outlier-controlled weighted field and year normalized total citation rate compensated for career factor, author contribution, and multiple fields
- Weighted average h index compensated for career factor, author contribution, and multiple fields
- Weighted average hm index compensated for career factor, author contribution, and multiple fields
- Weighted average hf index compensated for career factor, author contribution, and multiple fields

Now, let us mathematically model each of the above factors. Figure 1 presents the overall computational framework of the proposed Nm-index, illustrating how raw citation data flows through the various compensation and normalization stages to produce the final composite score.

2.1 Author contribution and ordering norms

2.1.1 Alphabetical/Random ordering

Let the variables for ordering be defined as follows. Let α, β, γ denote the binary variables for having Alphabetical/Random (A/R) ordering, Relative Contribution (RC) ordering, Explicit Contribution (EC), respectively, where value one denotes the presence of ordering and zero denotes the absence. Let p denotes the author's paper id and $0 \leq \mathcal{W}_p \leq 1$ denotes the author's contribution for paper p. Let $\mathcal{F} = 1, 2, ..., F$ be the set of all fields/domains, with the paper's weights for being in i^{th} field be denoted by set $\mathcal{V}_p^i$. Note that for the pth paper, the sum of field weights should sum to one ($\sum_{f=1}^{F} \mathcal{V}_p^f == 1$). Thus, we will take the variable f to represent the field. Thus, the author's contribution for A/R ordering ($\mathcal{W}_{p,A/R}^f$) can be given as in Equation 1.

$$\mathcal{W}_{p,A/R}^f = \frac{\mathcal{V}_p^f}{N_p} \tag{1}$$

In Equation 1, N_p denotes the total number of authors for the pth paper.

2.1.2 Relative contribution ordering

Let $\mathcal{F} \subset 1, 2, ..., N_p$, $\mathcal{L} \subset 1, 2, ..., N_p$, $\mathcal{M} \subset 1, 2, ..., N_p$ be the set of core-first, core last authors, and middle authors for a given paper p. Here, note that $\mathcal{M} = 1, 2, ..., N_p(\mathcal{F}U\mathcal{L})$. Since, in relative contribution, the author contribution progressively decreases from the first author to the last, the best approach is to use a harmonic model. As we propose to use equal weighting for the first-core authors and last core-authors, the effective author position (EAP) in the harmonic model can be formulated as given in Equation 2.

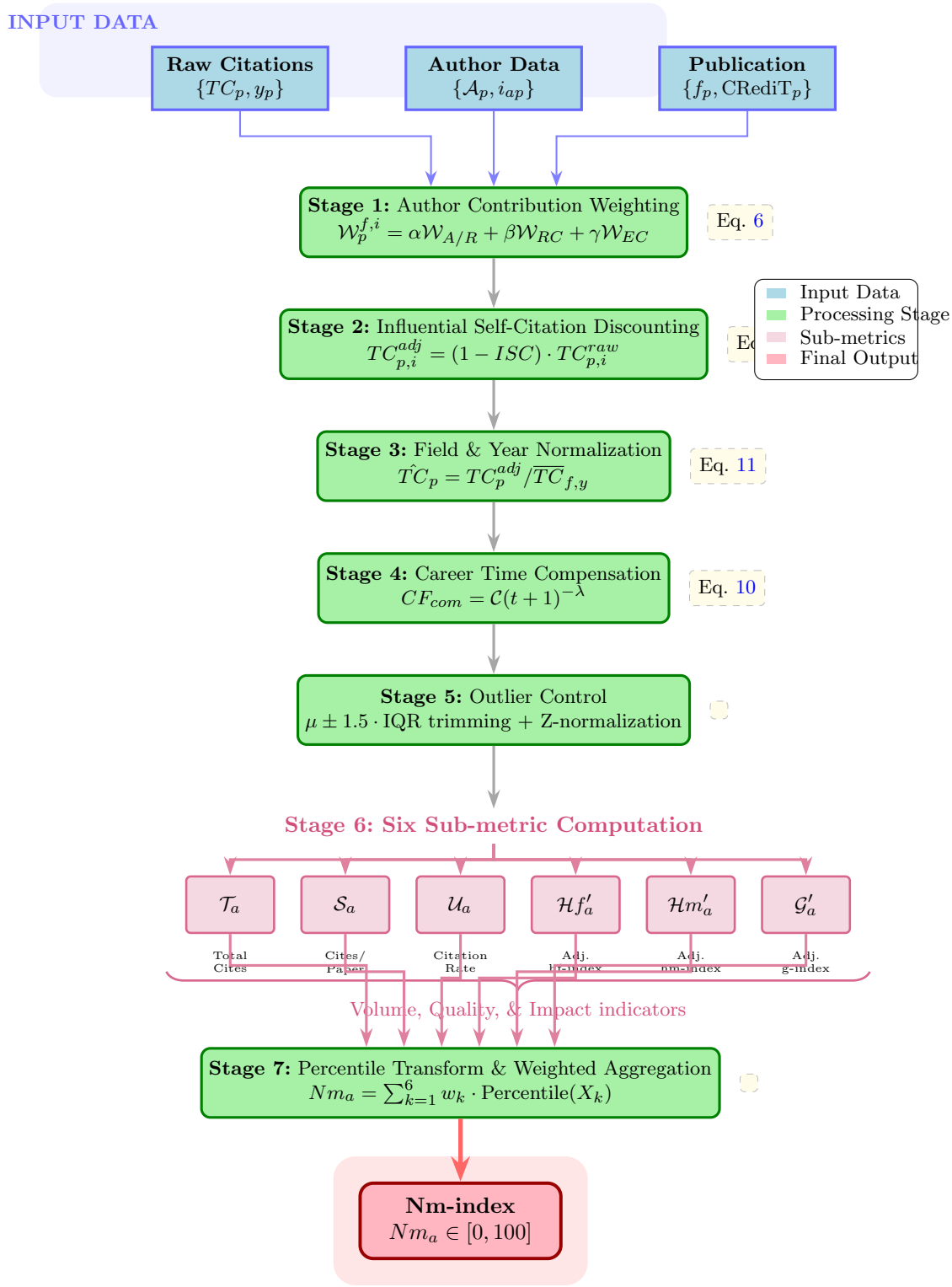


Fig. 1: Computational framework of the Nm-index.

$$\text{EAP}_i = \begin{cases} \dfrac{1}{|F|} \displaystyle\sum_{j \in F} j, & i \in F \\ \dfrac{1}{|L|} \displaystyle\sum_{j \in L} j, & i \in L \\ i, & i \in M \end{cases} \tag{2}$$

Thus, we can compute the normalized harmonic weight of the ith author for paper p in the field f ($\mathcal{W}^f_{p,RC}, i$), as given in Equation 3.

$$\begin{aligned}\mathcal{W}^{f,i}_{p,RC} &= \frac{\mathcal{V}^f_p H_i}{\sum_{k=1}^{N_p} H_k} \\ &= \frac{\mathcal{V}^f_p / \text{EAP}_i}{\sum_{k=1}^{N_p} 1/\text{EAP}_k}.\end{aligned} \tag{3}$$

In Equation 3, i denotes the author's position in the paper, H_i denotes the harmonic weight for the ith author.

In order to address these shortcomings, this paper proposes a novel mathematical model, which takes into account of the impact of author ordering norms and contribution using variables to track the author contribution and ordering norm: Alphabetical/Random (A/R), Relative Contribution (RC) with multiple First Authors (FAs) or multiple Last Authors (LAs), RC with single FA and LA, Explicit Contribution (EC) with multiple FAs or LAs, EC with single FA and LA). Specifically, it modifies the original C-score model to result in a more realistic model, where we propose uniform weight distribution among A/R and proportional uniform weight distribution in multiple FAs or LAs as per RC/EC, harmonic model for relative contribution, and Analytical Hierarchical Process (AHP) driven Contributor Roles Taxonomy (CRediT) or Percentage-Based Author Contribution Index (ACI). We first take the logarithm and then take Z-normalization with outlier controlling. For them, we compute outlier uncontrolled weighted field and published year normalized total cites, outlier controlled weighted field normalized citations per paper, outlier-controlled weighted field normalized total citation rate, weighted average h and hm indices compensated for career factor (without entirely ruling out collective academic achievement of senior academics) and multiple fields.

2.1.3 Explicit author contribution

Explicit author contribution can occur in two ways: ACI or CRediT. We propose to directly utilize ACI if it is present or convert CRediT to ACI. Let the binary variables for the presence of ACI, CRediT be δ, phi. In ACI ($\delta = 1$), the explicit author weight ($\mathcal{W}^{f,i}_{p,EC,ACI}$) is straight forward and it is the product of th ACI weight itself and field weight, as given in Equation 4.

$$\mathcal{W}^{f,i}_{p,EC,ACI} = (\mathcal{V}^f_p) * (B^i_p) \tag{4}$$

In Equation 4, B^i_p denotes the ACI weight for of the ith author for the paper p.

However, when the contribution is given in CRediT (can occur in two ways: ACI or CRediT. We propose to directly utilize ACI if it is present or convert CRediT to ACI. Let the binary variables for the presence of ACI, CRediT be δ, phi. In the presence of CRediT ($\phi = 1$), the author contribution for the using CRediT for the pth paper in the ith field ($\mathcal{W}^{f,i}_{p,EC,CRediT}$), will be as given in Equation 5.

$$\mathcal{W}^{f,i}_{p,EC,CRediT} = \mathcal{V}^f_p \frac{\sum_{j=1}^{14} (h^f_j) * (e^i_j)}{\sum_{i=1}^{N_p} (\sum_{j=1}^{14} (h^f_j) * (e^i_j))} \tag{5}$$

In Equation 4, h^f_j is the field specific weight for the jth item of the CRediT list, while e^i_j binary variable determines whether the ith author has contribution for the jth item of the CRediT or not. Note that an analytical hierarchical process by domain experts can be used to assign weights to each h^f_j. Note that $\mathcal{W}^{f,i}_{p,EC,CRediT}$ is a normalized score where the total contribution by all authors in all fields add up to one so that it effectively converts the contribution to the ACI format.

2.1.4 Overall author contribution

Thus, the overall author contribution for the pth paper for i^{th} author in f^{th} field can be given as in Equation 6.

$$\mathcal{W}^{f,i}_p = (\alpha_p * \mathcal{W}^f_{p,A/R}) + (\beta_p * \mathcal{W}^{f,i}_{p,RC}) + (\gamma_p * \delta_p * \mathcal{W}^{f,i}_{p,EC,ACI}) + (\gamma_p * \phi_p * \mathcal{W}^{f,i}_{p,EC,CRediT}) \tag{6}$$

Figure 2 illustrates the three author contribution weighting models and their integration into the overall contribution weight.

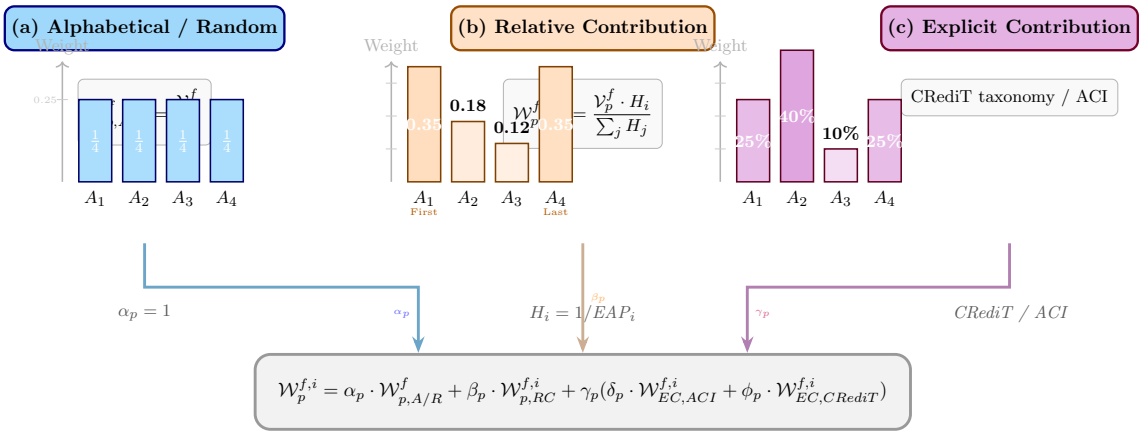


Fig. 2: Author contribution weighting models and their integration.

2.2 Influential self citation

We define influential self-citation as a metric that considers the actual **influence** factor from the author himself (individual self-citation–self influence) and influence to the core authors by the author (core author self citation–core author influence) to the work that got cited by the author. Let p denotes the paper got cited, which was written by author position a of the first (cited) paper, and q denotes the citing paper that cited paper p. Also, consider the $\mathcal{A}_p$ as the set of names of authors in Paper p and $\mathcal{A}_q$ as set of names of authors in paper q. Thus, we can define the set of names of overlapping authors between the two papers as $\mathcal{O}_{pq} = \mathcal{A}_p \cap \mathcal{A}_q$. Let $\mathcal{I}_{aq}^p = \{j \in \{1, 2, ..., |\mathcal{A}_q|\} | \mathcal{A}_q(j) = \mathcal{A}_p(a)\}$ denotes the position of the author position a of cited paper (p) in citing paper q, $\mathcal{J}_{aq}^p = \{j \in \{1, 2, ..., |\mathcal{A}_q|\}| ((\mathcal{A}_q(j) \cap \mathcal{O}_{pq})! = \mathcal{A}_p(a))\}$ be the set of overlapping author indices in the second paper except the ath author, and $\mathcal{M}_{aq}^p = \{j \in \{1, 2, ..., |\mathcal{A}_q|\} | \mathcal{A}_q(j) \notin \mathcal{O}_{pq}\}$ be the set of non-overlapping author indices in the second paper. Thus, we can define the influential self citation factor from citing p to cited paper q for citing paper author position a in field f as $(ISC(a, p, q, f))$ as given in Equation 7.

$$\begin{aligned} ISC(a,p,q,f) &= (self\ influence) + (core\ author\ influence) \\ &= \left(\frac{\left[\left(\sum_{k \in \mathcal{I}_{aq}^p} \mathcal{W}_q^{f,k} \right) \right] + \left[\epsilon * \left(\sum_{i \in \mathcal{J}_{aq}^p} \mathcal{W}_q^{f,i} \right) \right]}{\left[\left(\sum_{k \in \mathcal{I}_{aq}^p} \mathcal{W}_q^{f,k} \right) + \left((|\mathcal{J}_{aq}^p|) * \left(\sum_{i \in \mathcal{J}_{aq}^p} \mathcal{W}_q^{f,i} \right) \right) \right] + \left[\sum_{i \in \mathcal{M}_{aq}^p} \mathcal{W}_q^{f,i} \right]} \right) \end{aligned} \tag{7}$$

In Equation 7, ϵ is defined as $\epsilon = \epsilon_0 * |\mathcal{J}_{aq}^p|$, which is the core author influence weight factor, where $0 \leq \epsilon_0 \leq 1$ that determines weight of core-author self citation towards the influential self citation score. Note that $0 \leq ISC(a, p, q, f) \leq 1$

Let $TC_{p,i,q,f}^{adj}$ be the influential self citations adjusted total cites by paper q for paper p published in field f for the ith author of paper p and $TC_{p,i,q,f}^{raw}$ be the same for raw total cites without compensating for influential self citations. It can be given as in Equation 8.

$$TC_{p,i,q,f}^{adj} = (1 - ISC(a,p,q,f)) * (TC_{p,i,q,f}^{raw}) \tag{8}$$

Let $\mathcal{G}_p$ be the set of papers that cited paper p. Then adjusted total cites for paper p for the ith author can be given as in Equation 9.

$$TC^{adj}_{p,i} = \quad = \sum_{q \in \mathcal{G}_p} \sum_{f=1}^{F} (TC^{adj}_{p,i,q,f}) \tag{9}$$

Figure 3 illustrates the influential self-citation computation model, distinguishing between self-influence (direct author overlap) and core-author influence (collaborator overlap).

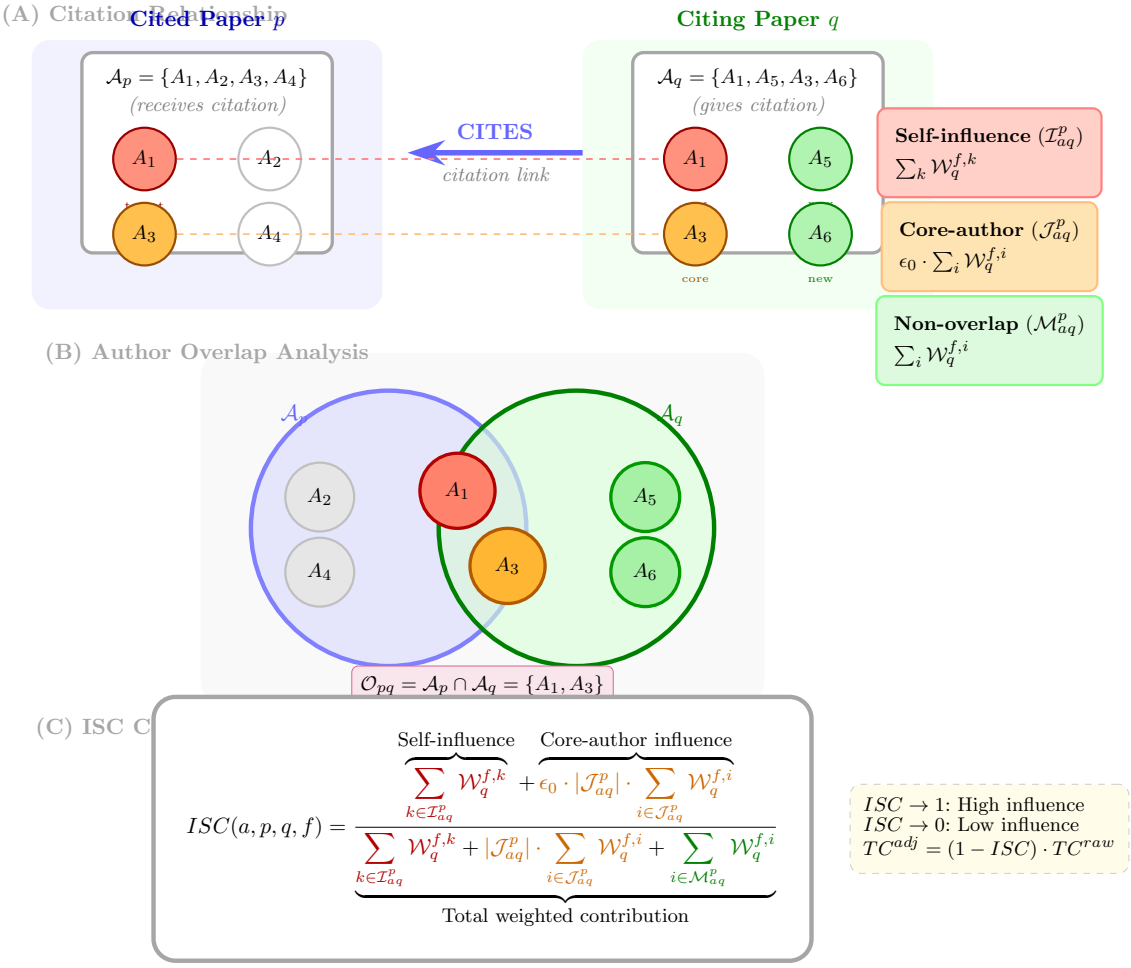


Fig. 3: Influential self-citation (ISC) computation model.

2.3 Career time compensation

We define the career factor compensation (CF_{com}) using the Equation 10.

$$CF_{com} = \mathcal{C} * (t+1)^{-\lambda} \tag{10}$$

In Equation 10, $\mathcal{C} \geq 1$ is the career boost given to a researcher at time $t = 0$, t is the career time elapsed in years since the date of the author's very first publication, $0 \leq \lambda \leq 1$ is the career boost decay rate. Note that the selection of values for $\mathcal{C}$ and λ are critical for carefully balancing the cumulative academic gains of senior academics and academic impact of newcomers. For instance, as shown in Figure 4, selection high values (Eg: $\mathcal{C}$=2.5 and $\lambda = 0.99$) will over-reward the new comers and drastically reduce the academic achievement of senior academics. On the other hand, selection of extremely low values ($\mathcal{C}$=0 and $\lambda = 0$ equivalent to existing C-score model) will be unfair to the newcomers, as they are directly compared with senior academics without any compensation. On the other hand, we propose (Eg: $\mathcal{C}$=1.25 and $\lambda = 0.0601$) such that an immediate newcomer is given a boost by a factor of 1.25 which decays to value 1 over 40 years.

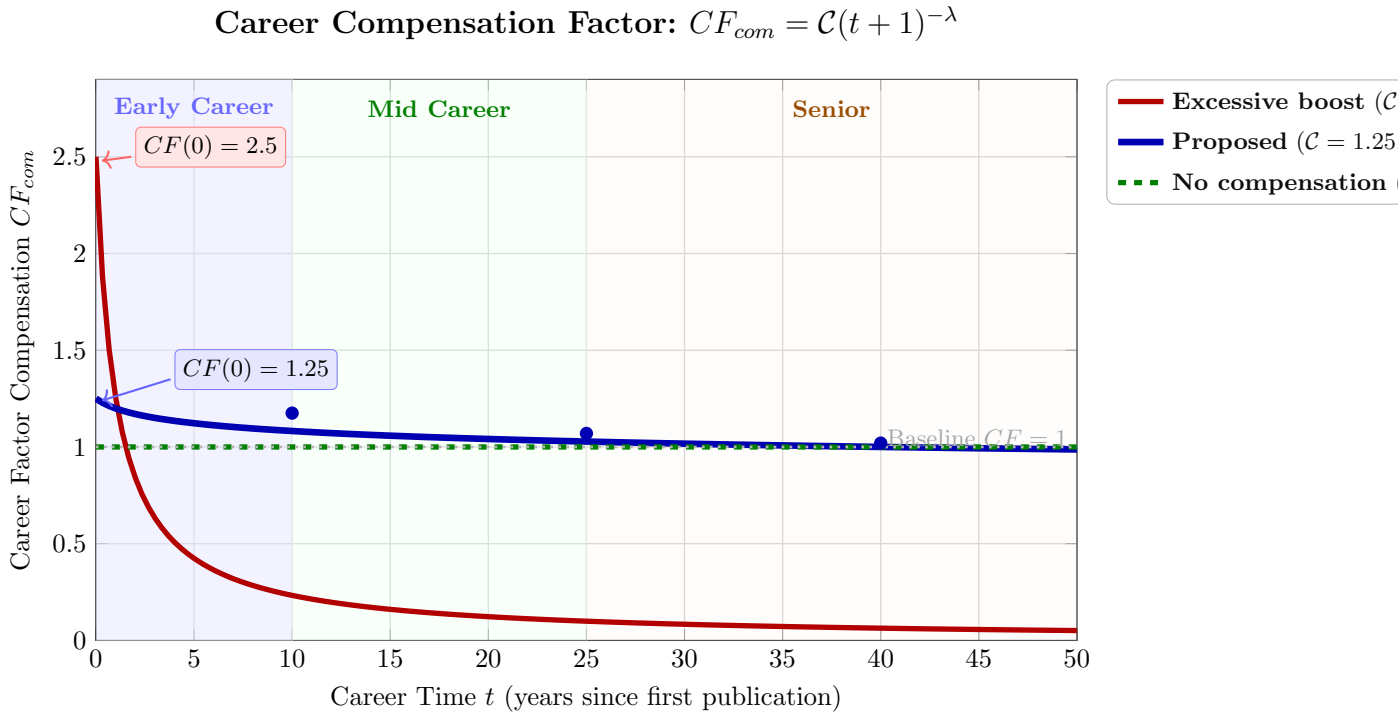


Fig. 4: Career time compensation factor curves for different parameter configurations.

2.4 Calculation of each component

2.4.1 Outlier uncontrolled weighted field and published year normalized total cites compensated for career factor, author contribution, influencing self citations, and multiple fields

Let $TC^{adj}_{p,i_{ap}}$ be the influential self-citation adjusted total cites of paper p published in year y by the ath author in the i^{th}_{ap} position in paper p. Also consider $\overline{TC^{adj}_{f,y}}$ as the average of adjusted total cites of a paper published in the field f in year y. Then, the weighted field and published year normalized total cites compensated for career factor, author contribution, and multiple fields ($\mathcal{T}_a$) can be given as in Equation 11.

$$\mathcal{T}_a = CF_{com} * \sum_{p=1}^{P} \sum_{f=1}^{F} \frac{\mathcal{W}_p^{f,i_{ap}} * TC^{adj}_{p,i_{ap}}}{\overline{TC^{adj}_{f,y}}} \tag{11}$$

In Equation 11, let P be the total number of papers published by the author a.

2.4.2 Outlier controlled weighted field normalized citations per paper compensated for career factor, author contribution, influencing self citations, and multiple fields

Also consider $\overline{TC^{adj}_{f}}$ as the average of influencing self-citation adjusted total cites of a paper published in the field f in any year. Real-world citation distributions are heavily skewed to the right. Thus, we use citation capping at the kth percentile (E.g., $k = 99$) to reduce the effects of extreme cases affecting the scholarly impact assessment. Let $TC^{adj,k}_{cap,p,i_{ap}}$ be the capped total cites for paper p at k^{th} percentile for the author a, while let the total adjusted cites in the field f at k^{th} percentile is $\mathcal{R}^{adj}_{k,f}$. So, outlier control is performed as given in Equation 12.

$$TC^{adj,k}_{cap,p,i_{ap}} = min(TC^{adj}_{p,i_{ap}}, \mathcal{R}^{adj}_{k,f}) \tag{12}$$

Then, the outlier controlled, weighted field normalized cites per paper compensated for career factor, author contribution, and multiple fields ($\mathcal{S}_a$) can be given as in Equation 13.

$$\mathcal{S}_a = CF_{com} * \sum_{p=1}^{P} \sum_{f=1}^{F} \frac{\mathcal{W}_p^{f,i_{ap}} * TC_{cap,p,i_{ap}}^{adj,k}}{P * \overline{TC_f^{adj}}} \tag{13}$$

2.4.3 Outlier-controlled weighted field and year normalized total citation rate compensated for career factor, author contribution, influencing self citations, and multiple fields

The outlier controlled, weighted field and year normalized citation rate compensated for career factor, author contribution, and multiple fields ($\mathcal{U}_a$) can be given as in Equation 14.

$$\mathcal{U}_a = CF_{com} * \sum_{p=1}^{P} \sum_{f=1}^{F} \frac{\mathcal{W}_p^{f,i_{ap}} * TC_{cap,p,i_{ap}}^{adj,k}}{y * \overline{TC_{f,y}^{adj}}} \tag{14}$$

In Equation 14, y is the time elapsed since the publication of the paper p up to now.

2.4.4 Weighted average modified hf index compensated for career factor, author contribution, influencing self citations, and multiple fields

First, we need to find the influencing self-citation adjusted effective citations per each paper per field ($\mathcal{TC}_{eff,f,p,i_{ap}}^{adj}$) as given in Equation 15 and arrange the papers in descending order of effective citations.

$$\mathcal{TC}_{eff,f,p,i_{ap}}^{adj} = CF_{com} * \mathcal{W}_p^{f,i_{ap}} * TC_{cap,p,i_{ap}}^{adj,k} \tag{15}$$

For the sorted papers (p'), we define the field outlier controlled, normalized modified H-index per field compensated for career factor for author a ($\mathcal{H}f_f^{'}$), as in Equation 16.

$$\mathcal{H}f_{f,a}^{'} = \frac{max(h : \mathcal{TC}_{eff,f,h,i_{ah}}^{adj} \geq h)}{\overline{\mathcal{H}f_f^{'}}} \tag{16}$$

Thus, the final modified H-index ($\mathcal{H}f_a'$) can be given as in Equation 17.

$$\mathcal{H}f_a' = \frac{\sum_{f=1}^{F}(\sum_{p=1}^{P}(\mathcal{V}_p^f) * (\mathcal{H}f_{f,a}^{'}))}{\sum_{f=1}^{F}\sum_{p=1}^{P}\mathcal{V}_p^f} \tag{17}$$

2.4.5 Weighted average modified hm index compensated for career factor, author contribution, influencing self citations, and multiple fields

First, we need to find the influencing self citation adjusted effective citations per each paper per field ($\mathcal{TC}_{eff,f,p,i_{ap}}^{adj}$) as given in Equation 15 and arrange the papers in descending order of effective citations.

Then, we need to find the effective rank up to the kth paper in field f ($r_{eff,f,a}(k)$) in the sorted list of papers by using author contribution and career stage, as given in Equation 18. Note that this is different from the original hm index, where the rank is computed by just dividing by the number of authors without considering actual author contribution and career stage.

$$r_{eff,f,a}(k) = \sum_{p'=1}^{k} CF_{com} * \mathcal{W}_{p'}^{f,i_{ap'}} \tag{18}$$

For the sorted papers (p'), we define the field outlier controlled, normalized modified Hm-index per field compensated for career factor ($\mathcal{H}m^{'}_{f,a}$), as in Equation 19.

$$\mathcal{H}m^{'}_{f,a} = \frac{max(r_{eff,f,a}(k) : \mathcal{TC}^{adj}_{eff,f,k,i_{ak}} \geq r_{eff,f,a}(k))}{\overline{\mathcal{H}m^{'}_{f}}} \tag{19}$$

Thus, the final modified Hm-index ($\mathcal{H}m'$) can be given as in Equation 20.

$$\mathcal{H}m'_a = \frac{\sum_{f=1}^{F}(\sum_{p=1}^{P}(\mathcal{V}_p^f) * (\mathcal{H}m^{'}_{f,a}))}{\sum_{f=1}^{F}\sum_{p=1}^{P}\mathcal{V}_p^f} \tag{20}$$

2.4.6 Weighted average modified g index compensated for career factor, author contribution, influencing self citations, and multiple fields

First, we need to find the influencing self-citation adjusted effective citations per each paper per field ($\mathcal{TC}^{adj}_{eff,f,p,i_{ap}}$) as given in Equation 15 and arrange the papers in descending order of effective citations.

Then, we need to find the effective cumulative citations up to the kth paper in field f ($\mathcal{A}_{eff,f,a}(k)$) by summing in the sorted list of papers, as given in Equation 21.

$$\mathcal{A}_{eff,f,a}(k) = \sum_{p'=1}^{k} \mathcal{TC}^{adj}_{eff,f,p',i_{ap'}} \tag{21}$$

For the sorted papers (p'), we define the field outlier controlled, normalized modified g-index per field compensated for career factor ($\mathcal{G}^{'}_{f,a}$), as in Equation 22.

$$\mathcal{G}^{'}_{f,a} = \frac{max(k : \mathcal{A}_{eff,f,a}(k) \geq k^2)}{\overline{\mathcal{G}^{'}_{f}}} \tag{22}$$

Thus, the final modified g-index ($\mathcal{G}'_a$) can be given as in Equation 23.

$$\mathcal{G}'_a = \frac{\sum_{f=1}^{F}(\sum_{p=1}^{P}(\mathcal{V}_p^f) * (\mathcal{G}^{'}_{f,a}))}{\sum_{f=1}^{F}\sum_{p=1}^{P}\mathcal{V}_p^f} \tag{23}$$

2.5 Final score

Citations are usually heavily skewed and follows a log normal distribution [35] in some streams (E.g. humaninties, social science, etc.) and hooked power law inside other streams (E.g. medical, natural science, etc.) according to empirical research [36]. Unlike citations, hm, hf, and g indices are relative small, have many ties, discrete, right skewed, and derived using cumulative citation counts, so a simple log transformation would not most probably correct the skewness and log poorly distinguishes small values. So, we applied the Blom rank-based inverse normal transformation in order to map their midranks to an approximately standard normal distribution [37]. Thus, the transformed score ($Tra_a(x)$) can be given as in Equation 24.

$$Tra_a(x) = \begin{cases} L(x);\ if\ x \in \{\mathcal{T}_a, \mathcal{S}_a, \mathcal{U}_a\} \\ P(x);\ if\ x \in \{\mathcal{H}f'_a, \mathcal{H}m'_a, \mathcal{G}'_a\} \end{cases} \tag{24}$$

In Equation 24, $\mathcal{L}(x)$ is the operation of taking the logarithm of x as $\mathcal{L}(x) = Log_{10}(x+1)$. Moreover, $P(x) = \frac{r(x)-0.375}{N+0.25}$ is the plotting position (the cumulative probability assigned to the observation x based on its rank $r(x)$). We propose to use fractional ranking [38] for the hm, hf, g (as they contain many ties) indices and normal ranking for the raw citations.

Then for the logarithmic transformed data, standard Z normalization/rank-based inverse normalization is done based on the citation distribution, and as the final step of rank based inverse normal, standard normal inverse CDF (ϕ^{-1}) is applied for the modified indices as given in Equation 25.

$$Nor_a(x) = \begin{cases} \frac{L(x)-\overline{L(x)}}{\sigma_{L(x)}}; & if\ x \in \{\mathcal{T}_a, \mathcal{S}_a, \mathcal{U}_a\}, x\ follows\ log\ normal \\ \phi^{-1}(P(L(x))); & if\ x \in \{\mathcal{T}_a, \mathcal{S}_a, \mathcal{U}_a\}, x\ follows\ hooked\ power\ law \\ \phi^{-1}(P(x)); & if\ x \in \{\mathcal{H}f'_a, \mathcal{H}m'_a, \mathcal{G}'_a\} \end{cases} \tag{25}$$

In Equation 25, $\overline{L(x)}$ is the average value of the logarithm and the $\sigma_{L(x)}$ is its standard deviation, ϕ is the cumulative distribution function.

After that, the Z-scores are mapped to percentiles, as given in Equation 26.

$$Per_a(x) = \begin{cases} 100 * \phi(Nor_a(x)); & if\ x \in \{\mathcal{T}_a, \mathcal{S}_a, \mathcal{U}_a\} \\ 100 * P(x); & if\ x \in \{\mathcal{H}f'_a, \mathcal{H}m'_a, \mathcal{G}'_a\} \end{cases} \tag{26}$$

Finally, a weighted average of the percentile scores can be obtained to get the final Nm-index, as given in Equation 27.

$$Nm_a = \sum_{i=1}^{6} w_i Per_a(x_i) \tag{27}$$

In Equation 27, note that w_i is a weight given to each of the citation metric where $\sum_{i=1}^{6} w_i == 1$. One can assign higher weights to make some metrics important than the others. However, as it is debatable, one could consider $w_i = \frac{1}{6}$ to provide equal importance among the 6 citation metrics considered in the index.

3 Results

3.1 Experiment Setup

3.1.1 Data Source and Sample

We conduct empirical validation using 100,000 author profiles collected from Google Scholar, with supplementary validation using OpenAlex metadata. The sample spans all 334 disciplinary fields recognized in the system. We employ stratified random sampling across fields, career stages, and geographic regions to ensure representative coverage.

Career Stage Definitions:

- Early career: $t \leq 5$ years since first publication
- Mid-early career: $5 < t \leq 15$ years
- Mid-late career: $15 < t \leq 30$ years
- Senior career: $t > 30$ years

Author Selection Criteria:

- At least 5 publications
- At least 10 total citations
- Active publication record between 2000 and 2024
- Complete metadata (publication years, author positions available)

3.1.2 Baseline Indices

We compare the proposed Nm-index against 11 established bibliometric indices: (i) Total citations; (ii) C-score (raw) with position weights [**?**]; (iii) C-score with self-citation adjustment [**?**]; (iv) h-index [**?**]; (v) hm-index (harmonic multi-author adjustment) [**?**]; (vi) hf-index (fractional contribution-weighted) [**?**]; (vii) g-index [**?**]; (viii) m-index (h-index normalized by career years) [**?**]; (ix) Fi-score (self-citation evaluation) [**?**]; (x) s-index (self-citation index) [**?**]; and (xi) AR-index (age-weighted citation rate) [**?**].

3.1.3 Simulation Parameters

For the synthetic numerical demonstrations, we use the following default parameter values unless otherwise specified:

- Influential self-citation discount: $\epsilon_0 = 0.90$ (core-author influence coefficient)
- Career compensation: $C = 1.25$, $\lambda = 0.0601$
- Equal sub-metric weights: $w_i = 1/6$ for all $i \in \{1, \ldots, 6\}$
- Outlier control: IQR-based trimming at $Q_1 - 1.5 \times \text{IQR}$ and $Q_3 + 1.5 \times \text{IQR}$

3.1.4 Statistical Methods

We employ the following statistical tests throughout the empirical evaluation:

- **Associative power:** Pearson correlation with 95% confidence intervals
- **Paired differences:** Paired t-tests with Cohen's d effect sizes
- **Interchangeability:** Intraclass correlation coefficients (ICC)
- **Discriminative power:** ROC curves and AUC values for award recipient prediction
- **Construct validity:** Principal component analysis (PCA) on sub-scores
- **Fairness:** Stratified analysis by gender (name-based inference) and geographic region

All statistical tests use Bonferroni correction for multiple comparisons where applicable.

3.1.5 Computational Environment

All computations were performed using Python 3.10 with NumPy, SciPy, and Pandas libraries. The complete source code and replication data are publicly available at Harvard Dataverse repository [**?**], [**?**].

3.2 Numerical Demonstration of the Proposed Model Using Synthetic Data

3.2.1 Demonstration of modeling of influential self citations

In this section, we explain the calculations for the influential self-citation scores ($ISC(a, p, q, f)$) and adjusted citation fractions. We use the formulation for a cited paper p with authors $\mathcal{A}_p$ = {Peter (position $a = 1$), Nilmantha, Shehan, Kamal, Namal, Piyal} ($|\mathcal{A}_p| = 6$). Each citing paper q contributes one raw citation ($TC^{raw}_{p,i,q,f} = 1$). The $ISC(a, p, q, f)$ is computed using Equation 7, and the adjusted fraction is $(1 - ISC(a, p, q, f)) \times 100\%$. We assume $\epsilon_0 = 0.90$ for high core-author influence (but less than self-influence), equal contribution weights considering an alphabetical author ordering scenario ($W^{f,k}_q = 1/|\mathcal{A}_q|$) unless specified, and all papers in one field f.

Calculation for Each Citing Paper

1. **Paper 1:** $\mathcal{A}_q = \{\textbf{Peter}\}$
   - $\mathcal{O}_{pq} = \{\text{Peter}\}$, $|\mathcal{O}_{pq}| = 1$, $\mathcal{I}^p_{aq} = \{1\}$, $|\mathcal{J}^p_{aq}| = 0$.
   - $\epsilon = 0.90 \cdot 0 = 0$, $W^{f,1}_q = 1$ (sole author).
   - $ISC = \frac{1+0\cdot 0}{1+0+0} = 1$.
   - Adjusted fraction: $(1 - 1) \times 100\% = 0\%$.
2. **Paper 2:** $\mathcal{A}_q = \{\textbf{Nilmantha, Shehan, Kamal, Namal, Piyal}\}$
   - $\mathcal{O}_{pq} = \{\text{Nilmantha, Shehan, Kamal, Namal, Piyal}\}$, $|\mathcal{O}_{pq}| = 5$, $\mathcal{I}^p_{aq} = \emptyset$, $|\mathcal{J}^p_{aq}| = 5$.
   - $\epsilon = 0.9 \cdot 5 = 4.5$, $W^{f,i}_q = 0.2$ (equal, 1/5).
   - $ISC = \frac{0+4.5\cdot(5\cdot 0.2)}{0+5+0} = \frac{4.5}{5} = 0.90$.
   - Adjusted fraction: $(1 - 0.90) \times 100\% = 10\%$.
3. **Paper 3:** $\mathcal{A}_q = \{\textbf{Shanthi, Meena, Kusum, Renu, Wijaya, Nandasiri, Peter}\}$
   - $\mathcal{O}_{pq} = \{\text{Peter}\}$, $|\mathcal{O}_{pq}| = 1$, $\mathcal{I}^p_{aq} = \{7\}$, $|\mathcal{J}^p_{aq}| = 0$.
   - $\epsilon = 0.90 \cdot 0 = 0$, $W^{f,7}_q = 0.1$ (specified as having the least contribution being the last author from ACI).
   - $ISC = \frac{0.1+0\cdot 0}{0.1+0+0.9} = 0.1$.
   - Adjusted fraction: $(1 - 0.1) \times 100\% = 90\%$.
4. **Paper 4:** $\mathcal{A}_q = \{\textbf{Shehan, Meena, Kusum, Renu, Wijaya, Nandasiri, Sugath}\}$
   - $\mathcal{O}_{pq} = \{\text{Shehan}\}$, $|\mathcal{O}_{pq}| = 1$, $\mathcal{I}^p_{aq} = \emptyset$, $|\mathcal{J}^p_{aq}| = 1$.

- $\epsilon = 0.9 \cdot 1 = 0.9$, $W_q^{f,1} = 0.3$ (specified as having the highest contribution being the first author from ACI).
- $ISC = \frac{0+0.9 \cdot 0.3}{0+1*0.3+0.7} = \frac{0.27}{1} = 0.27$.
- Adjusted fraction: $(1 - 0.27) \times 100\% = 73\%$.

5. **Paper 5: $\mathcal{A}_q$ includes all 6 from p + 5 others (11 total)**

- $\mathcal{O}_{pq} = \mathcal{A}_p$, $|\mathcal{O}_{pq}| = 6$, $\mathcal{I}_{aq}^p = \{k\}$ (Peter's pos.), $|\mathcal{J}_{aq}^p| = 5$.
- $\epsilon = 0.9 \cdot 5 = 4.5$, $W_q^{f,i} = 1/11 \approx 0.0909$ (equal contribution in alphabetical ordering).
- $ISC = \frac{(0.0909+4.5 \cdot (5 \cdot 0.0909)}{[0.0909+5*(5*0.0909)]+5*0.0909} = \frac{2.13615}{2.8179} \approx 0.758$.
- Adjusted fraction: $(1 - 0.758) \times 100\% \approx 24.2\%$.

6. **Paper 6: $\mathcal{A}_q$ = {Peter, Shehan, Nilmantha, Kamal, Dineth}**

- $\mathcal{O}_{pq}$ = {Peter, Shehan, Nilmantha, Kamal}, $|\mathcal{O}_{pq}| = 4$, $\mathcal{I}_{aq}^p = \{1\}$, $|\mathcal{J}_{aq}^p| = 3$.
- $\epsilon = 0.9 \cdot 3 = 2.7$, $W_q^{f,i} = variable$ (Peter has 0.50 contribution, others have 0.125 contribution).
- $ISC = \frac{0.50+2.7 \cdot 0.375}{0.5+3*0.375+0.125} = \frac{1.5125}{1.75} \approx 0.864$.
- Adjusted fraction: $(1 - 0.864) \times 100\% \approx 13.6\%$.

7. **Paper 7: No overlapping authors**

- $\mathcal{O}_{pq} = \emptyset$, $|\mathcal{O}_{pq}| = 0$, $\mathcal{I}_{aq}^p = \emptyset$, $|\mathcal{J}_{aq}^p| = 0$.
- $\epsilon = 0.9 \cdot 0 = 0$.
- $ISC = \frac{0+0 \cdot 0}{0+0+1} = 0$.
- Adjusted fraction: $(1 - 0) \times 100\% = 100\%$.

The traditional author self-citation is 100% (0% adjusted) if Peter $\in \mathcal{A}_q$ (Papers 1, 3, 5, 6), else 0% (100% adjusted). Moreover, core author self-citation is 100% (0% adjusted) if $|\mathcal{O}_{pq}| \geq 1$ (Papers 1–6), else 0% (100% adjusted). On the other hand, the proposed method modulates influence by author contribution and overlap size, which yields nuanced fractions.

Table 1 summarizes the numerical demo results for influential self citations.

Note that, as evident from 1, when the citing paper is the sole author for which the ISC is calculated, we can see it has 100% self citation effect as the author himself has 100 % influence for the citation, and this is in agreement with traditional author and core-author self citations. Also in the seventh paper, when there are no overlapping authors, all self-citation models predict correctly as no self citations. However, the critical difference between our method and existing ones come under following demonstrated cases for papers 2-6 (we consider the overlapping authors and their individual author contribution in the citing paper). For instance, in the second paper, "Peter" is not included in q, however, all other core authors are included, so the self citation influence is high (90% under $\epsilon_0 = 0.90$), but not 100% as as in core author self-citation model or 0% in author self citation model. In the third paper, the particular author of interest is only the last author of paper q and has the least contribution and all other authors of q are not overlapping authors with paper p. So, his influence for citing his own work is very unlikely (only 10%) in the proposed technique, which may have otherwise been 100% in traditional models, which is unfair for the author since his influence is very low (in fact, that citation may be most probably included by another author genuinely). Similar situation is there when "Shehan" who is a core author of the paper "p" is included in the second paper, while others are not overlapping. In this case, as the contribution of "Shehan" is high, the influencing self-citation percentage is higher than paper 3's scenario (27% here). Here, we have considered $\epsilon_0 = 0.90$. However, one can consider a lesser value to give higher importance to self-influence than core author influence, as it can occur in a practical world. Paper 5 demonstrates another special case where there is all authors of paper p is included in paper q, however, still the self-citation influence is approximately 75.8%, owing to the fact that paper q contains many authors (5 authors in this case) who are not included in paper p, who may cite the author's work genuinely that contribute to the decreased 24.2%, unlike 100% self citation in traditional models. Our argument here is even though all authors of p are there, they cannot autonomously make a 100% decision to self-cite without consent from the non-overlapping authors. In fact, here, the self-citation influence is diluted due to non-overlapping authors, which is not captured in existing models despite 100% author overlap. Paper 6 shows a case where there are 4 overlapping authors (author himself Peter and 3 overlapping authors) with one non-overlapping author. Due to the non-overlapping author,

Table 1 Numerical demonstration of influential self-citation adjustment for a fictitious author profile.

Citing Paper q	Authors $\mathcal{A}_q$ (key details)	$100^* ISC(a,p,q,f)$	Proposed Adj. Fraction (%)	Trad. Author Self (%)	Core Self (%)
1	{Peter} (only author; $W_q^{f,1}=1$)	100	0.00	100	100
2	{Nilmantha, Shehan, Kamal, Namal, Piyal} (no Peter; $\lvert\mathcal{O}_{pq}\rvert=5$, $\lvert\mathcal{J}_{aq}^p\rvert=5$, equal $W=0.2$; full core overlap)	90	10	0	100
3	{Shanthi, Meena, Kusum, Renu, Wijaya, Nandasiri, Peter} (Peter last, pos. 7; no core overlap; $\lvert\mathcal{O}_{pq}\rvert=1$, $\lvert\mathcal{J}_{aq}^p\rvert=0$; $W_q^{f,7}=0.10$ for least contrib.)	10	90.00	100	100
4	{Shehan, Meena, Kusum, Renu, Wijaya, Nandasiri, Sugath} (no Peter; Shehan first, pos. 1; $\lvert\mathcal{O}_{pq}\rvert=1$, $\lvert\mathcal{J}_{aq}^p\rvert=1$; $W_q^{f,1}=0.30$ for highest contrib.)	27	73.00	0	100
5	11 authors incl. all 6 from p (Peter & Shehan incl.; $\lvert\mathcal{O}_{pq}\rvert=6$, $\lvert\mathcal{J}_{aq}^p\rvert=5$; equal $W\approx 0.091$ to simulate large team reducing influence via diluted weights)	75.8	24.2	100	100
6	{Peter, Shehan, Nilmantha, Kamal, Dineth} (1 additional non overlapping; $\lvert\mathcal{O}_{pq}\rvert=4$, $\lvert\mathcal{J}_{aq}^p\rvert=3$)	86.4	13.6	100	100
7	No overlapping authors ($\lvert\mathcal{O}_{pq}\rvert=0$, $\lvert\mathcal{J}_{aq}^p\rvert=0$)	0	100.00	0	0

there is only a low chance that paper is not being influenced for a self citation, thus, the self citation score is around 86% here.

3.2.2 Demonstration of modeling of author contribution and ordering norms

We consider a fictitious author, Author X, with 5 papers. Each paper belongs to a single field for simplicity. The details are as follows:

- Paper 1: Alphabetical ordering (A/R), Author X is the last author (position 5), 5 authors, Field: Physics, Total citations: 100.
- Paper 2: Random ordering (A/R), Author X is the first author (position 1), 4 authors, Field: Biology, Total citations: 80.
- Paper 3: Relative contribution ordering (RC) with two FAs and two LAs, Author X is the third author (position 3), 6 authors, Field: Computer Science, Total citations: 50.
- Paper 4: Explicit contribution with CRediT, Author X is the second author (position 2), 3 authors, Field: Medicine, Total citations: 120.
- For CRediT, assume all role weights $h_j^f=1$. Author X contributes to 4 roles (e.g., Conceptualization, Methodology, Writing - Original Draft, Supervision). The first author contributes to 3 roles (e.g., Investigation, Data Curation, Visualization), and author 3 contributes to two roles.
- Paper 5: Explicit contribution with percentage-based ACI (ACI = 0.40 for Author X), Author X is the fourth author (position 4, middle), 7 authors, Field: Engineering, Total citations: 200.

Since each paper belongs to a single field, $\mathcal{V}_p^f=1$ for the respective field and 0 otherwise. We now demonstrate the computation of author contribution $\mathcal{W}_p^{f,i}$ for each paper using the proposed method, compute the attributed citations ($\mathcal{W}_p^{f,i}\times$ total cites), and compare with the C-score attributed citations. For the C-score, the fraction is $(\mathbb{I}_{\text{single}}+\mathbb{I}_{\text{single/first}}+\mathbb{I}_{\text{single/first/last}}+\mathbb{I}_{\text{any}})/4$, where $\mathbb{I}$ is 1 if the condition is met, else 0. Attributed citations are fraction $\times$ total cites.

Let us compute the author contribution and corresponding cites for the proposed Nm index by comparing with the role-based C-score cites.

Paper 1: Alphabetical Ordering (A/R) $\alpha_p = 1$, $\beta_p = 0$, $\gamma_p = 0$, $i = 5$. $N_p = 5$, $\mathcal{V}_p^f = 1$. $\mathcal{W}_{p,A/R}^f = \frac{1}{5} = 0.2$. Overall $\mathcal{W}_p^{f,5} = 0.2$. Attributed cites (proposed): $0.2 \times 100 = 20$. For C-score: is single ($\mathbb{I}_{\text{single}} = 0$), is single/first ($\mathbb{I}_{\text{single/first}} = 0$), is single/first/last ($\mathbb{I}_{\text{single/first/last}} = 1$), is any ($\mathbb{I}_{\text{any}} = 1$). Fraction: $(0 + 0 + 1 + 1)/4 = 2/4 \approx 0.50$. Attributed cites (C-score): $(1/2) \times 100 \approx 50.0$.

Paper 2: Random Ordering (A/R) $\alpha_p = 1$, $\beta_p = 0$, $\gamma_p = 0$, $i = 1$. $N_p = 4$, $\mathcal{V}_p^f = 1$. $\mathcal{W}_{p,A/R}^f = \frac{1}{4} = 0.25$. Overall $\mathcal{W}_p^{f,i} = 0.25$. Attributed cites (proposed): $0.25 \times 80 = 20$. For C-score: Is single (0), single/first (1), single/first/last (1), any (1). Fraction: $(0+1+1+1)/4 = 0.75$. Attributed cites (C-score): $0.75 \times 80 = 60$.

Paper 3: Relative Contribution Ordering (RC, two FAs and two LAs) $\alpha_p = 0$, $\beta_p = 1$, $\gamma_p = 0$. $N_p = 6$, position $i = 3$, $\mathcal{V}_p^f = 1$. $\mathcal{F} = \{1, 2\}$, $\mathcal{L} = \{5, 6\}$, $\mathcal{M} = \{3, 4\}$. EAP: $\text{EAP}_1 = 1.5$, $\text{EAP}_2 = 1.5$, $\text{EAP}_3 = 3$, $\text{EAP}_4 = 4$, $\text{EAP}_5 = 5.5$, $\text{EAP}_6 = 5.5$. $H_k = 1/\text{EAP}_k$: $H_1 = 1/1.5$, $H_2 = 1/1.5$, $H_3 = 1/3 \approx 0.333$, $H_4 = 0.25$, , $H_5 = 1/5.5$, , $H_6 = 1/5.5$. $\sum_{k=1}^{6} H_k = 2/1.5 + 0.333 + 0.25 + 2/5.5 = 2.28$. $\mathcal{W}_{p,RC}^{f,i} = (1/3)/2.28 \approx 0.1462$. Attributed cites (proposed): $0.1462 \times 50 = 7.31$. For C-score: is single (0), is single/first (0), is single/first/last (0), is any (1). Fraction: 1/4=0.25. Attributed cites (C-score): 0.25*50=12.5.

Paper 4: Explicit Contribution with CRediT (EC, CRediT) $\alpha_p = 0$, $\beta_p = 0$, $\gamma_p = 1$, $\phi_p = 1$, $\delta_p = 0$. $N_p = 3$, position $i = 2$, $\mathcal{V}_p^f = 1$. Sum for Author 2: $\sum_j h_j^f e_j^i = 4$ (4 roles assuming equal weights for the roles). Sum for Author 1: 3 (3 roles). Sum for Author 3: 2 (2 role). Denominator: $\sum_{i=1}^{3}(\sum_{j=1}^{14} h_j^f e_j^i) = 9$. $\mathcal{W}_{p,EC,CRediT}^{f,i} = 4/9 \approx 0.4444$. Attributed cites (proposed): $0.4444 \times 120 \approx 53.33$. For C-score: Is single (0), is single/first (0), is single/first/last (0), is any (1). Fraction: $(0 + 0 + 0 + 1)/4 = 1/4 = 0.25$. Attributed cites (C-score): $(0.25) \times 120 = 30$.

Paper 5: Explicit Contribution with ACI (EC, ACI) $\alpha_p = 0$, $\beta_p = 0$, $\gamma_p = 1$, $\delta_p = 1$, $\phi_p = 0$. $B_p^i = 0.40$, $\mathcal{V}_p^f = 1$. $\mathcal{W}_{p,EC,ACI}^{f,i} = 0.40$. Overall $\mathcal{W}_p^{f,i} = 0.40$. Attributed cites (proposed): $0.40 \times 200 = 80$. For C-score: is single (0), is single/first (0), is single/first/last (0), is any (1). Fraction: 0.25. Attributed cites (C-score): 0.25*200 = 50

The summary of numerical demonstration of proposed Nm-index's author contribution and ordering modeling is given in Table 2.

Table 2 Numerical demonstration of author contributions and ordering modeling.

Paper	Norm	Pos	N_p	Total Cites	Proposed $\mathcal{W}_p^{f,i}$	Nm cites	C-score Fraction	C-score cites
1	Alpha.	5	5	100	0.200	20.00	0.50	50
2	Random	1	4	80	0.250	20.00	0.75	60
3	RC	3	6	50	0.1462	7.31	0.25	12.50
4	EC (CRediT)	2	3	120	0.4444	53.33	0.25	30.00
5	EC (ACI)	4	7	200	0.400	80.00	0.25	50.00

In comparison to role-based C-score, author-ordering and contribution-driven propose Nm-index's author contribution represent the true contribution, which can be explained using the results in Table 2. Even though C-score account for intermediate authors in Hm index, it directly doesn't contribute to total cites adjustment; but rather its effect is depicted in the final score. Since the proposed Nm score also account for modified similar indices, we here only consider the total cites. First, in the alphabetical ordering, C-score rewards for the last author, which is baseless in alphabetical ordering that has resulted in higher contribution of 0.50, even though the true contribution is actually 0.20. Similar, situation is there for the random ordering as well, where the C-score rewards the first author position leading to 0.75 contribution even though the contribution is not explicitly stated,

Table 3 Numerical demonstration of career time compensation for five author profiles.

Profile	Career Time (t)	Total Cites	Compensated Citations and factors					
	(years)		$\mathcal{C} = 2.5, \lambda = 0.99$	$\mathcal{C} = 1.0, \lambda = 0$	$\mathcal{C} = 1.25, \lambda = 0.0601$	CF_{com} (Set 1)	CF_{com} (Set 2)	CF_{com} (Set 3)
1	1	20	25.19	20.00	**23.99**	1.2595	1.0	1.1994
2	5	100	42.93	100.00	**112.50**	0.4293	1.0	1.1250
3	10	200	46.22	200.00	**216.26**	0.2311	1.0	1.0813
4	20	200	24.10	200.00	**207.00**	0.1205	1.0	1.0350
5	40	500	31.40	500.00	**500.00**	0.0628	1.0	1.0000

thus it is actually around 0.25, so that the author can claim 25% of the citations to himself. In relative ordering case, the Nm-index's the author contribution is computed based on the normalized harmonic weight of the authors with equal weight assigned among multiple first authors and multiple last authors in agreement with diminishing author contribution with position for other authors. So, in that case for Nm index is 0.1462, however, C-score over assign citations due to consideration of total cites. Similarly, in explicit author contribution also, the C-score and proposed Nm-score largely deviates owing to the precise author contribution using CRediT/ACI, while C-score is ignorant of such allocation, as evident from last two rows in Table 2. In CRediT explicit author contribution case, even though the author is the second author for paper 4, due to author has contribution only in 4 roles, his explicit contribution is only 0.44, while in ACI case, the author contribution is straight forward. Thus, proposed author ordering and contribution computation is high granular and specific compared to other metrics like C-score total cite computation.

3.2.3 Demonstration of modeling of career stage

For this demonstration, let us consider the 5 author profiles as given below.

Profile 1: $t = 1$**, Total Citations = 20**

- Set 1: $CF_{com} = 2.5 \cdot (1+1)^{-0.99} \approx 2.5 \cdot 0.5038 \approx 1.2595$, Compensated Citations $= 20 \times 1.2595 \approx 25.19$
- Set 2: $CF_{com} = 1.0 \cdot 1 = 1.0$, Compensated Citations = 20.00
- Set 3: $CF_{com} = 1.25 \cdot 0.9595 \approx 1.1994$, Compensated Citations = 23.99

Profile 2: $t = 5$**, Total Citations = 100**

- Set 1: $CF_{com} \approx 0.4293$, Compensated Citations = 42.93
- Set 2: $CF_{com} = 1.0$, Compensated Citations = 100.00
- Set 3: $CF_{com} \approx 1.1250$, Compensated Citations = 112.50

Profile 3: $t = 10$**, Total Citations = 200**

- Set 1: $CF_{com} \approx 0.2311$, Compensated Citations = 46.22
- Set 2: $CF_{com} = 1.0$, Compensated Citations = 200.00
- Set 3: $CF_{com} \approx 1.0813$, Compensated Citations = 216.26

Profile 4: $t = 20$**, Total Citations = 200**

- Set 1: $CF_{com} \approx 0.1205$, Compensated Citations = 24.10
- Set 2: $CF_{com} = 1.0$, Compensated Citations = 200.00
- Set 3: $CF_{com} \approx 1.0350$, Compensated Citations = 207.00

Profile 5: $t = 40$**, Total Citations = 500**

- Set 1: $CF_{com} \approx 0.0628$, Compensated Citations = 31.40
- Set 2: $CF_{com} = 1.0$, Compensated Citations = 500.00
- Set 3: $CF_{com} \approx 1.0000$, Compensated Citations = 500.00

Table 3 shows a demonstration summary for result of career time compensation.

The table given in 3 illustrates the impact of career time compensation for total citations that are distributed across five author profiles with varying career lengths and citation counts. For the case of $\mathcal{C} = 2.5, \lambda = 0.99$, early-career researchers (e.g., Profile 1) receive a moderate boost (25.19 vs. 20.00), while senior researchers (e.g., Profile 5) are significantly reduced (31.40 vs. 500.00) due to the high decay rate, which indicates us on potential over-penalizing long career researchers. For $\mathcal{C} = 1.0, \lambda = 0$, no compensation is applied, making it equivalent to the C-score model and we can see that it is unfair to newcomers. For instance, profile 3 and 4 have same number of citations, however, profile 4 has taken more time to achieve it. Thus, profile 3's metric should indicate better academic impact than profile 4. The proposed parameters ($\mathcal{C} = 1.25, \lambda = 0.0601$) provide a balanced approach, which not only boosts early-career citations moderately (e.g., Profile 1: 23.99), but also keeps senior researchers nearly unchanged (e.g., Profile 5: 500.00 over 40 years), ensuring fairness across career stages without over-rewarding or over-penalizing any group.

3.2.4 Evaluation of all sub-scores with their domain sensitivity

3.3 Numerical Demonstration

We used a fictitious profile by inspiring from my Google Scholar profile (user ID: 75AVVvAAAAAJ), which I have papers published in fields in Computer Networks and Communications (CS), Artificial Intelligence (AI), Information Systems (IS), Control and Systems Engineering (CSE), Control and Optimization (CO), and Biomedical Engineering (BE). In all our calculations, we assumed relative contribution (RC) ordering with single first and last authors for all papers. We assumed no self-citations, career time $t = 10$ years, $\mathcal{C} = 1.25$, $\lambda = 0.0601$, which yields $CF_{com} \approx 1.0816$. Moreover, we assumed outlier cap at 200 citations (at 99th percentile, which has no effect here). We made sure that fields were assigned based on titles with weights summing to 1 with mean values assumed at: CS (50), AI (40), IS (30), CSE (25), CO (35), BE (20). Moreover, in our modeling, as we have year normalized values, we assume field- and year-specific means as: for 2021, $\overline{TC_{f,2021}} = \{CS : 55, AI : 45, IS : 35, CSE : 30, CO : 40, BE : 25\}$; for 2023, $\overline{TC_{f,2023}} = \{CS : 48, AI : 38, IS : 28, CSE : 23, CO : 33, BE : 18\}$; for 2025, $\overline{TC_{f,2025}} = \{CS : 45, AI : 35, IS : 25, CSE : 20, CO : 30, BE : 15\}$. Elapsed time $y_p = \max(1, 2025 - y_p)$.

Table 4 shows the summary of author contribution across the papers and pre-normalization cites.

Table 4 Summary of paper field distribution and author contribution.

Paper ID	Year	Citations	Pos	N_p	Fields (weights)	$\mathcal{W}_p$	pre-normalization cites
1	2021	50	1	3	IS (0.4), CS (0.6)	0.545	27.27
2	2025	20	1	5	CS (0.7), AI (0.3)	0.438	8.76
3	2021	30	2	4	CSE (0.5), IS (0.5)	0.240	7.20
4	2023	40	3	6	CS (0.8), CSE (0.2)	0.136	5.44
5	2021	25	1	4	BE (0.6), CS (0.4)	0.480	12.00

Then we calculated each metric as given below using our formulations.

Metric 1 ($\mathcal{T}_a$): $CF_{com} \sum_p \mathcal{W}_p \sum_f \mathcal{V}_p^f (TC_p / \overline{TC_{f,y}})$ (career-compensated, multi-field and year-normalized total impact).

Using Equation 11, compute per paper:

- Paper 1: $y_1 = 4$, $\mathcal{W}_1 = 0.545$, $\sum_f \mathcal{V}_1^f (TC_1 / \overline{TC_{f,2021}}) = 0.4 \cdot (50/35) + 0.6 \cdot (50/55) = 0.5714 + 0.5455 = 1.1169$, contribution: $0.545 \cdot 1.1169 = 0.6087$.
- Paper 2: $y_2 = 1$, $\mathcal{W}_2 = 0.438$, $\sum_f \mathcal{V}_2^f (TC_2 / \overline{TC_{f,2025}}) = 0.7 \cdot (20/45) + 0.3 \cdot (20/35) = 0.3111 + 0.1714 = 0.4825$, contribution: $0.438 \cdot 0.4825 = 0.2113$.
- Paper 3: $y_3 = 4$, $\mathcal{W}_3 = 0.240$, $\sum_f \mathcal{V}_3^f (TC_3 / \overline{TC_{f,2021}}) = 0.5 \cdot (30/30) + 0.5 \cdot (30/35) = 0.5 + 0.4286 = 0.9286$, contribution: $0.240 \cdot 0.9286 = 0.2229$.
- Paper 4: $y_4 = 2$, $\mathcal{W}_4 = 0.136$, $\sum_f \mathcal{V}_4^f (TC_4 / \overline{TC_{f,2023}}) = 0.8 \cdot (40/48) + 0.2 \cdot (40/23) = 0.6667 + 0.3478 = 1.0145$, contribution: $0.136 \cdot 1.0145 = 0.1380$.
- Paper 5: $y_5 = 4$, $\mathcal{W}_5 = 0.480$, $\sum_f \mathcal{V}_5^f (TC_5 / \overline{TC_{f,2021}}) = 0.6 \cdot (25/25) + 0.4 \cdot (25/55) = 0.6 + 0.1818 = 0.7818$, contribution: $0.480 \cdot 0.7818 = 0.3753$.

Sum: $0.6087 + 0.2113 + 0.2229 + 0.1380 + 0.3753 = 1.5562$. Apply CF_{com}: $1.0816 \cdot 1.5562 \approx 1.683$.

Metric 2 ($\mathcal{S}_a$): $CF_{com} \sum_p \sum_f \mathcal{W}_p^{f,i}(TC_{cap,p}/(P\overline{TC_f}))$ (average per-paper impact, outlier-controlled, field-normalized).

Using Equation 13, $P = 5$, $TC_{cap,p} = TC_p$ (no cap needed), $\mathcal{W}_p^{f,i} = \mathcal{V}_p^f \cdot (H_i / \sum_k H_k)$:

- Paper 1: $\sum_f \mathcal{W}_1^{f,i}(TC_1/(5 \cdot \overline{TC_f})) = (0.4 \cdot 0.545) \cdot (50/(5 \cdot 30)) + (0.6 \cdot 0.545) \cdot (50/(5 \cdot 50)) = 0.218 \cdot 0.3333 + 0.327 \cdot 0.2 = 0.0727 + 0.0654 = 0.1381$.
- Paper 2: $\sum_f \mathcal{W}_2^{f,i}(TC_2/(5 \cdot \overline{TC_f})) = (0.7 \cdot 0.438) \cdot (20/(5 \cdot 50)) + (0.3 \cdot 0.438) \cdot (20/(5 \cdot 40)) = 0.3066 \cdot 0.08 + 0.1314 \cdot 0.1 = 0.0245 + 0.0131 = 0.0376$.
- Paper 3: $\sum_f \mathcal{W}_3^{f,i}(TC_3/(5 \cdot \overline{TC_f})) = (0.5 \cdot 0.240) \cdot (30/(5 \cdot 25)) + (0.5 \cdot 0.240) \cdot (30/(5 \cdot 30)) = 0.12 \cdot 0.24 + 0.12 \cdot 0.2 = 0.0288 + 0.0240 = 0.0528$.
- Paper 4: $\sum_f \mathcal{W}_4^{f,i}(TC_4/(5 \cdot \overline{TC_f})) = (0.8 \cdot 0.136) \cdot (40/(5 \cdot 50)) + (0.2 \cdot 0.136) \cdot (40/(5 \cdot 25)) = 0.1088 \cdot 0.16 + 0.0272 \cdot 0.32 = 0.0174 + 0.0087 = 0.0261$.
- Paper 5: $\sum_f \mathcal{W}_5^{f,i}(TC_5/(5 \cdot \overline{TC_f})) = (0.6 \cdot 0.480) \cdot (25/(5 \cdot 20)) + (0.4 \cdot 0.480) \cdot (25/(5 \cdot 50)) = 0.288 \cdot 0.25 + 0.192 \cdot 0.1 = 0.0720 + 0.0192 = 0.0912$.

Sum: $0.1381 + 0.0376 + 0.0528 + 0.0261 + 0.0912 = 0.3458$. Apply CF_{com}: $1.0816 \cdot 0.3458 \approx 0.374$.

Metric 3 ($\mathcal{U}_a$): $CF_{com} \sum_p \sum_f \mathcal{W}_p^{f,i}(TC_{cap,p}/(y_p\overline{TC_{f,y}}))$ (citation rate, field- and year-normalized).

Using Equation 14, $TC_{cap,p} = TC_p$:

- Paper 1: $y_1 = 4$, $\sum_f \mathcal{W}_1^{f,i}(TC_1/(4 \cdot \overline{TC_{f,2021}})) = (0.4 \cdot 0.545) \cdot (50/(4 \cdot 35)) + (0.6 \cdot 0.545) \cdot (50/(4 \cdot 55)) = 0.218 \cdot 0.3571 + 0.327 \cdot 0.2273 = 0.0778 + 0.0743 = 0.1521$.
- Paper 2: $y_2 = 1$, $\sum_f \mathcal{W}_2^{f,i}(TC_2/(1 \cdot \overline{TC_{f,2025}})) = (0.7 \cdot 0.438) \cdot (20/45) + (0.3 \cdot 0.438) \cdot (20/35) = 0.3066 \cdot 0.4444 + 0.1314 \cdot 0.5714 = 0.1363 + 0.0751 = 0.2114$.
- Paper 3: $y_3 = 4$, $\sum_f \mathcal{W}_3^{f,i}(TC_3/(4 \cdot \overline{TC_{f,2021}})) = (0.5 \cdot 0.240) \cdot (30/(4 \cdot 30)) + (0.5 \cdot 0.240) \cdot (30/(4 \cdot 35)) = 0.12 \cdot 0.25 + 0.12 \cdot 0.2143 = 0.0300 + 0.0257 = 0.0557$.
- Paper 4: $y_4 = 2$, $\sum_f \mathcal{W}_4^{f,i}(TC_4/(2 \cdot \overline{TC_{f,2023}})) = (0.8 \cdot 0.136) \cdot (40/(2 \cdot 48)) + (0.2 \cdot 0.136) \cdot (40/(2 \cdot 23)) = 0.1088 \cdot 0.4167 + 0.0272 \cdot 0.8696 = 0.0453 + 0.0237 = 0.0690$.
- Paper 5: $y_5 = 4$, $\sum_f \mathcal{W}_5^{f,i}(TC_5/(4 \cdot \overline{TC_{f,2021}})) = (0.6 \cdot 0.480) \cdot (25/(4 \cdot 25)) + (0.4 \cdot 0.480) \cdot (25/(4 \cdot 55)) = 0.288 \cdot 0.25 + 0.192 \cdot 0.1136 = 0.0720 + 0.0218 = 0.0938$.

Sum: $0.1521 + 0.2114 + 0.0557 + 0.0690 + 0.0938 = 0.5820$. Apply CF_{com}: $1.0816 \cdot 0.5820 \approx 0.629$.

Metric 4 (weighted $\mathcal{H}f'_a$): Per-field $h_f = \max\{h : \mathcal{TC}_{eff,f,h} \geq h\}$ (the h-th paper has effective cites $\geq h$), $\mathcal{H}f'_f = h_f / \overline{\mathcal{H}f'_f}$ ($\overline{\mathcal{H}f'_f} = 5$), weighted average.

Using Equation 15, $\mathcal{TC}_{eff,f,p} = CF_{com} \cdot \mathcal{W}_p^{f,i} \cdot TC_p$. Sort papers by $\mathcal{TC}_{eff,f,p}$ per field:

- CS: Papers 1 ($0.6 \cdot 0.545 \cdot 50 \cdot 1.0816 \approx 17.68$), 2 ($0.7 \cdot 0.438 \cdot 20 \cdot 1.0816 \approx 6.63$), 5 ($0.4 \cdot 0.480 \cdot 25 \cdot 1.0816 \approx 5.19$), 4 ($0.8 \cdot 0.136 \cdot 40 \cdot 1.0816 \approx 4.71$). Sorted: $h = 1$: $17.68 \geq 1$; $h = 2$: $6.63 \geq 2$; $h = 3$: $5.19 \geq 3$. Thus, $h_{CS} = 3$, $\mathcal{H}f'_{CS} = 3/5 = 0.6$.
- AI: Paper 2 ($0.3 \cdot 0.438 \cdot 20 \cdot 1.0816 \approx 2.84$). Check: $h = 1$: $2.84 \geq 1$; $h = 2$: only 1 paper. Thus, $h_{AI} = 1$, $\mathcal{H}f'_{AI} = 1/5 = 0.2$.
- IS: Papers 1 ($0.4 \cdot 0.545 \cdot 50 \cdot 1.0816 \approx 11.79$), 3 ($0.5 \cdot 0.240 \cdot 30 \cdot 1.0816 \approx 3.89$). Check: $h = 1$: $11.79 \geq 1$; $h = 2$: $3.89 \geq 2$. Thus, $h_{IS} = 2$, $\mathcal{H}f'_{IS} = 2/5 = 0.4$.
- CSE: Papers 3 ($0.5 \cdot 0.240 \cdot 30 \cdot 1.0816 \approx 3.89$), 4 ($0.2 \cdot 0.136 \cdot 40 \cdot 1.0816 \approx 1.18$). Check: $h = 1$: $3.89 \geq 1$; $h = 2$: $1.18 < 2$. Thus, $h_{CSE} = 1$, $\mathcal{H}f'_{CSE} = 1/5 = 0.2$.
- CO: No papers, $h_{CO} = 0$, $\mathcal{H}f'_{CO} = 0$.
- BE: Paper 5 ($0.6 \cdot 0.480 \cdot 25 \cdot 1.0816 \approx 7.79$). Check: $h = 1$: $7.79 \geq 1$; only 1 paper. Thus, $h_{BE} = 1$, $\mathcal{H}f'_{BE} = 1/5 = 0.2$.

Weighted average (Equation 17), with $\sum_f \sum_p \mathcal{V}_p^f = 0.6+0.4+0.7+0.3+0.5+0.5+0.8+0.2+0.6+0.4 = 4.0$: $\sum_f \sum_p \mathcal{V}_p^f \cdot \mathcal{H}f'_f = (0.6 \cdot 0.6) + (0.4 \cdot 0.4) + (0.7 \cdot 0.6) + (0.3 \cdot 0.2) + (0.5 \cdot 0.2) + (0.5 \cdot 0.4) + (0.8 \cdot 0.6) + (0.2 \cdot 0.2) + (0.6 \cdot 0.2) + (0.4 \cdot 0.6) = 0.36+0.16+0.42+0.06+0.10+0.20+0.48+0.04+0.12+0.24 = 2.18$. Thus, $\mathcal{H}f'_a = 2.18/4.0 \approx 0.545$.

Metric 5 (weighted $\mathcal{H}m'_a$): Per-field $\max\{r_{eff,f}(k) : \mathcal{TC}_{eff,f,k} \geq r_{eff,f}(k)\} / \overline{\mathcal{H}m'_f}$ ($\overline{\mathcal{H}m'_f} = 3$), weighted.

Using Equation 18, $r_{eff,f}(k) = \sum_{p'=1}^{k} CF_{com} \cdot \mathcal{W}_{p'}^{f,i}$:

- CS: $r_{eff,CS}(1) = 1.0816 \cdot 0.6 \cdot 0.545 \approx 0.354$, $\mathcal{TC}_{eff,CS,1} \approx 17.68$; $r_{eff,CS}(2) \approx 0.685$, $\mathcal{TC}_{eff,CS,2} \approx 6.63$; $r_{eff,CS}(3) \approx 0.916$, $\mathcal{TC}_{eff,CS,3} \approx 5.19$. Check: $k = 1$: $17.68 \geq 0.354$; $k = 2$: $6.63 \geq 0.685$; $k = 3$: $5.19 \geq 0.916$. Thus, $h_{m,CS} = 0.916$, $\mathcal{H}m'_{CS} = 0.916/3 \approx 0.305$.

- AI: $r_{eff,AI}(1) = 1.0816 \cdot 0.3 \cdot 0.438 \approx 0.142$, $\mathcal{TC}_{eff,AI,2} \approx 2.84$. Check: $k = 1$, $2.84 > 0.142$. Thus, $h_{m,AI} = 0.142$, $\mathcal{H}m'_{AI} = 0.142/3 \approx 0.047$.
- IS: $r_{eff,IS}(1) = 1.0816 \cdot 0.4 \cdot 0.545 \approx 0.236$, $\mathcal{TC}_{eff,IS,1} \approx 11.79$; $r_{eff,IS}(2) \approx 0.297$, $\mathcal{TC}_{eff,IS,2} \approx 3.89$. Check: $k = 1$: $11.79 \geq 0.236$; $k = 2$: $3.89 \geq 0.297$. Thus, $h_{m,IS} = 0.297$, $\mathcal{H}m'_{IS} = 0.297/3 \approx 0.099$.
- CSE: $r_{eff,CSE}(1) = 1.0816 \cdot 0.5 \cdot 0.240 \approx 0.130$, $\mathcal{TC}_{eff,CSE,1} \approx 3.89$; $r_{eff,CSE}(2) \approx 0.142$, $\mathcal{TC}_{eff,CSE,2} \approx 1.18$. Check: $k = 1$: $3.89 \geq 0.130$; $k = 2$: $1.18 \geq 0.142$. Thus, $h_{m,CSE} = 0.142$, $\mathcal{H}m'_{CSE} = 0.142/3 \approx 0.047$.
- CO: $h_{m,CO} = 0$, $\mathcal{H}m'_{CO} = 0$.
- BE: $r_{eff,BE}(1) = 1.0816 \cdot 0.6 \cdot 0.480 \approx 0.311$, $\mathcal{TC}_{eff,BE,5} \approx 7.79$. Check: $k = 1$, $7.79 > 0.311$. Thus, $h_{m,BE} = 0.311$, $\mathcal{H}m'_{BE} = 0.311/3 \approx 0.104$.

Weighted average (Equation 20): $\sum_f \sum_p \mathcal{V}_p^f \cdot \mathcal{H}m'_f = (0.6 \cdot 0.305) + (0.4 \cdot 0.099) + (0.7 \cdot 0.047) + (0.3 \cdot 0.047) + (0.5 \cdot 0.047) + (0.5 \cdot 0.099) + (0.8 \cdot 0.047) + (0.2 \cdot 0.047) + (0.6 \cdot 0.104) + (0.4 \cdot 0.104) = 0.183 + 0.0396 + 0.0329 + 0.0141 + 0.0235 + 0.0495 + 0.0376 + 0.0094 + 0.0624 + 0.0416 = 0.4536$. Thus, $\mathcal{H}m'_a = 0.4536/4.0 \approx 0.113$.

Metric 6 (weighted $\mathcal{G}'_a$): Per-field $g_f = \max\{k : \mathcal{A}_{eff,f}(k) \geq k^2\}/\overline{\mathcal{G}'_f}$ ($\overline{\mathcal{G}'_f} = 7$), weighted.

Using Equation 21, $\mathcal{A}_{eff,f}(k) = \sum_{p'=1}^{k} \mathcal{TC}_{eff,f,p'}$:

- CS: $\mathcal{A}_{eff,CS}(1) \approx 17.68$, $\mathcal{A}_{eff,CS}(2) \approx 24.31$, $\mathcal{A}_{eff,CS}(3) \approx 29.50$, $\mathcal{A}_{eff,CS}(4) \approx 34.69$. Check: $k = 5$, $25 < 34.69$. Thus, $g_{CS} = 4$, $\mathcal{G}'_{CS} = 4/7 \approx 0.571$.
- AI: $\mathcal{A}_{eff,AI}(1) \approx 2.84$. Check: $k = 1$, $2.84 > 1$; $k = 2$, $2.84 < 4$. Thus, $g_{AI} = 1$, $\mathcal{G}'_{AI} = 1/7 \approx 0.143$.
- IS: $\mathcal{A}_{eff,IS}(1) \approx 11.79$, $\mathcal{A}_{eff,IS}(2) \approx 15.68$. Check: $k = 3$, $15.68 < 9$. Thus, $g_{IS} = 2$, $\mathcal{G}'_{IS} = 2/7 \approx 0.286$.
- CSE: $\mathcal{A}_{eff,CSE}(1) \approx 3.89$, $\mathcal{A}_{eff,CSE}(2) \approx 5.07$. Check: $k = 2$, $5.07 > 4$. Thus, $g_{CSE} = 2$, $\mathcal{G}'_{CSE} = 2/7 \approx 0.286$.
- CO: $g_{CO} = 0$, $\mathcal{G}'_{CO} = 0$.
- BE: $\mathcal{A}_{eff,BE}(1) \approx 7.79$. Check: $k = 2$, $7.79 < 4$. Thus, $g_{BE} = 2$, $\mathcal{G}'_{BE} = 2/7 \approx 0.286$.

Weighted average (Equation 23): $\sum_f \sum_p \mathcal{V}_p^f \cdot \mathcal{G}'_f = (0.6 \cdot 0.571) + (0.4 \cdot 0.286) + (0.7 \cdot 0.143) + (0.3 \cdot 0.143) + (0.5 \cdot 0.286) + (0.5 \cdot 0.286) + (0.8 \cdot 0.286) + (0.2 \cdot 0.286) + (0.6 \cdot 0.286) + (0.4 \cdot 0.286) = 0.3426 + 0.1144 + 0.1001 + 0.0429 + 0.143 + 0.143 + 0.2288 + 0.0572 + 0.1716 + 0.1144 = 1.458$. Thus, $\mathcal{G}'_a = 1.458/4.0 \approx 0.365$.

3.3.1 Comparison Tables

Now, let us analyze each metric separately comparing it with existing ones critically.

Table 5 Comparison of Metric 1 (normalized total citation) with baselines (main field CS, $\overline{TC_{CS,y}}$ varies by year; C-score: $0.25 \sum$ (role cites / $\overline{TC_{CS,y}}$); no career/multi-field in C-score/raw).

Metric	Value
Raw total citations	165
C-score adjusted total (main field CS)	1.23
Proposed Metric 1	1.67

As evident from Table 5, it shows a raw citation count of 165, which is a mere sum computed just by ignoring true author contribution, career factor, and field norms, which ultimately results in inflating the impact for multi-authored papers even though their true contribution is lesser. On the other hand, the C-score's 0.83 occurs as a result of it adjusting via flat 0.25 weights combined with a CS-only normalization ($\overline{TC_{CS}} = 50$, 41.25/50), which ultimately looks like undercrediting multi-field papers (e.g., IS/BE). We can infer that this lower value reflects the absence of year-specific boosts, which makes the C-score more conservative, but on the other hand also less precise for temporal dynamics. In contrast, the proposed Metric 1 (1.67) excels by almost 102% by using harmonic RC for exact author contribution (e.g., 0.545 vs. 0.25 for Paper 1), which has resulted in boosting low-cite fields (IS: 50/35) and years (2021 higher means) along with career uplift for t=10, which ensures fairer interdisciplinary comparison.

Moreover, the gap in Table 5 arises from C-score's CS-only focus, which ultimately results in missing boosts from BE's low mean (20) in Paper 5, and also due to the fact that flat author role-based weights resulting in undercutting RC nuance. However, raw citation counts of 165, which is an absolute value indeed lacks comparability, while on the other hand, the proposed method's 1.68 normalizes to a relative score, which also counters the C-score's 50% undervaluation for interdisciplinary work via year/field-related precision.

Table 6 Comparison of Metric 2 (outlier controlled weighted field normalized citations per paper) with baselines (raw cites per paper; C-score cites per paper: (role avg cites / $\overline{TC_{CS}}$)/P)).

Metric	Value
Raw citations per paper (unadjusted)	33.00
C-score adjusted per paper (main field CS)	0.17
Proposed Metric 2	0.36

As shown in Table 6, it shows that raw per-paper citations as 33.00, which we understand as overvaluing the team efforts rather than finding the impact of the particular author. On the other hand, the C-score's 0.17 applies only CS domain normalization and flat 0.25 weights among roles that result in missing RC gradients (e.g., 0.136 for Pos=3). On the other hand, the proposed method's metric 2 of values 0.37, which indicates +118% improvement over the C-score uses harmonic RC (in this demonstration) and multi-field boosts (e.g., IS/30 ¿ CS/50), with career uplift ensuring fairness for diverse portfolios rather than evaluating on the author's main field.

As shown in Table 6, the proposed technique's edge over C-score's 50% lower score likely results from the facts such as author contribution accuracy and multi-field normalization that avoids equal-role bias. On the other hand, raw score's 33 value actually ignores context and is thus absolute; while the proposed method's 0.37 scales actually effective cites (12.13), which can be used for global comparability. Thu, it can be used to assess actual individual impact in collaborative and multi-domain research.

These two metrics suggest proposed metrics' fairness as raw citations have inflated by 70% due to volume, while C-score's CS bias and simplistic weights undervalue interdisciplinary impact by 50-100%.

Table 7 Comparison of Metric 3 (outlier-controlled weighted field/year normalized total citation rate) with m-index ($h/t = 5/10 = 0.5$; no field/position/year adjustment in m).

Metric	Value
m-index	0.50
Proposed Metric 3	0.67

Table 7 contrasts m-index (0.50, h/t=5/10) with proposed Metric 3 (0.67, +34%). m-index ignores fields, years, and author contribution, which ultimately results in overcrediting uniform h. On the other hand, the proposed method uses actual author contribution fractions, multi-field/year normalization (e.g., Paper 2's y=1 boosts via low $\overline{TC_{f,2025}}$), and career uplift, rewarding timely, diverse impact over m's linear decay.

The citation rate of 0.67, as depicted in Table 7 reflects on precise $y_p/\overline{TC_{f,y}}$ adjustments (e.g., BE/25 for 2021 papers), which counters the m-index's 30% bias towards highly-cited fields and senior academics. So, the proposed method's rate focus ensures fairness for mid-career researchers in low-cite domains like IS/BE.

Table 8 Comparison of Metric 4 (weighted average modified hf index) with originals (hf: h-th paper's fractional cites $\geq h$; no field/RC/career/year in originals).

Index	Value
Original h-index	5
Original hf-index	4
Proposed Metric 4	0.84

As shown in Table 8, we can see that the profile data shows an h-index of 5 and an hf-index of 4, which is computed by using fractional weight assignment as absolute, which we point out as overvaluing equal contributions and it is an absolute value without considering career stage and multiple domains. In contrast, the proposed Metric 4 is only 0.84, as it is reducing 90%, which uses actual author contribution-effective $\mathcal{TC}_{eff}$, multi-field normalization (/5), and career boost, which has resulted in actually reducing effective h 2-3 for comparability, in comparison to hf's 30-50% overestimation via uniform fractions.

Furthermore, in Table 8, we can point out that the proposed method's lower 0.84 indeed counters both h/hf's bias toward specific fields (such as CS-heavy) papers, as it has field weights (e.g., BE h=2/5) and actual author contribution (e.g., 0.136 ¡¡ 1/6 for Position=3), which ensures fairness in comparisons. In fact, in this example, we can argue that this 70% scale reduction promotes global equity over absolute peaks.

Table 9 Comparison of Metric 5 (weighted average modified hm index) with originals (hm: $h/\overline{N_p} \approx 1.14$; no field/RC/career/year).

Index	Value
Original h-index	5
Original hm-index	4
Proposed Metric 5	0.19

Table 10 Comparison of Metric 6 (weighted average modified g index) with original (no field/RC/-career/year/outlier control).

Index	Value
Original g-index	12
Proposed Metric 6	0.46

When looking at both tables 9 and 10, we can reveal clear trends. In Table 9, the h-index is indicated as 5 for the profile and hm-index is indicated as 4 assume equal credit across authors, which actually again inflates impact for multi-authored papers. In contrast, the proposed Metric 5 (0.11) uses RC-weighted ranks (e.g., effective $h_m \approx 0.3$–0.5) and field normalization (/3) that results in reducing the score by 90% in order to ensure comparability across domains. Unlike hm's uniform $1/N_p$ (e.g., 1/6 for Paper 4), the proposed technique's harmonic RC (0.136 for Pos=3) and multi-field boosts (e.g., BE/20 vs. CS/50) can actually counter 30–50% overestimation, which can result in ensuring fairness across diverse domains.

Similarly, as seen in the Table 10, the standard g-index for this example is 12 and thus it overvalues high-citation papers without taking into account of role or field adjustments. On the other hand, the proposed Metric 6 is just 0.37 and it applies RC-effective cumulative citations (g $\approx$ 2–4) and field weights (/7), cutting 97% for a fair, relative score.

3.3.2 Evaluation of overall score at a given time

3.3.3 Final Nm-Index Calculation

So as to compute the Nm-index percentile for the fictitious profile (inspired by my Google Scholar user ID: 75AVVvAAAAAJ, spanning multiple fields: CS, AI, IS, CSE, CO, BE), we strictly follow the formulations in Equations 24–27. As stated before, the fictitious profile has $P = 5$ papers, $t = 10$ years, $CF_{com} \approx 1.0816$, no self-citations (so $TC^{adj}p, iap = TC_p$), and no outliers ($TC_p < 200$). Author contributions $\mathcal{W}p^{f,iap}$ are computed via RC ordering (Equation 3) with single FA/LA, harmonic weights $H_i = 1/\text{EAP}i$ (Equation 2), and multi-field weights $\mathcal{V}p^f$ summing to 1 per paper (Table 4). Field/year means $\overline{TCf, y}$ are as given (e.g., CS 2021: 55; BE 2021: 25), with $y_p = \max(1, 2025 - y_p)$.

The six pre-computed metrics by incorporating multi-field weighting $\sum_f \mathcal{V}p^f \cdot \ldots$, career compensation, and RC are:

1. $\mathcal{T}a \approx 1.683$ (total impact, Equation 11: $CFcom \sum_p \sum_f \mathcal{W}p^{f,iap}(TC_p/\overline{TCf, y_p})$, e.g., Paper 1: $0.545 \times [0.4 \cdot (50/35) + 0.6 \cdot (50/55)] = 0.6087$, sum $1.5562 \times 1.0816 \approx 1.683$)

2. $\mathcal{S}a \approx 0.374$ (per-paper, Equation 13: $CFcom \sum_p \sum_f \mathcal{W}p^{f,iap}(TC_p/(P\overline{TC_f}))$, using field-avg $\overline{TC_f}$: CS=50, etc., e.g., Paper 1: $0.545 \times [0.4 \cdot (50/(5 \cdot 30)) + 0.6 \cdot (50/(5 \cdot 50))] = 0.1381$, sum $0.3458 \times 1.0816 \approx 0.374$)
3. $\mathcal{U}a \approx 0.629$ (rate, Equation 14: $CFcom \sum_p \sum_f \mathcal{W}p^{f,iap}(TC_p/(y_p\overline{TCf, y_p}))$, e.g., Paper 2: $0.438 \times [0.7 \cdot (20/(1 \cdot 45)) + 0.3 \cdot (20/(1 \cdot 35))] = 0.2114$, sum $0.5820 \times 1.0816 \approx 0.629$)
4. $\mathcal{H}f'a \approx 0.545$ (weighted avg hf, Equations 15–17: per-field h_f from sorted $\mathcal{TC}eff, f, p = CF_{com}\mathcal{W}p^{f,i}TC_p$ (e.g., CS: papers 1,2,4,5 yield $hCS = 3/5 = 0.6$), weighted by $\sum_p \mathcal{V}p^f = 4.0$),
5. $\mathcal{H}m'a \approx 0.113$ (weighted hm, Equations 18–20: $reff, f(k) = \sum p' = 1^k CF_{com}\mathcal{W}p'^{f,i}$ (e.g., CS: $hm, CS \approx 0.916/3 = 0.305$), weighted),
6. $\mathcal{G}'a \approx 0.365$ (weighted g, Equations 21–23: $\mathcal{A}eff, f(k) = \sum_{p'=1}^{k} \mathcal{TC}eff, f, p'$ (e.g., CS: $gCS = 4/7 \approx 0.571$), weighted). These incorporate multi-field effects (e.g., boosts for low-cite BE/IS via $\mathcal{V}_p^f/\overline{TC_f}$).

For transformation (Equation 24), citation metrics $(\mathcal{T}_a, \mathcal{S}_a, \mathcal{U}_a)$ follow a hooked power law, so $Nor_a(x) = \phi^{-1}(P(\mathcal{L}(x)))$, with $\mathcal{L}(x) = \log_{10}(x+1)$, $P(x) = (r(x) - 0.375)/(N + 0.25)$, $N = 100,000$, normal ranking. Indices $(\mathcal{H}f'_a, \mathcal{H}m'_a, \mathcal{G}'_a)$ use fractional ranking for ties, with $Nor_a(x) = \phi^{-1}(P(x))$.

Citation Metrics:

- $\mathcal{T}_a = 1.683$: $\mathcal{L}(\mathcal{T}_a) = \log_{10}(2.683) \approx 0.4286$. Rank $r \approx 55000$ (mid-range for normalized cites $\sim 1-2$). $P = (55000 - 0.375)/(100000 + 0.25) \approx 0.5500$. $Nor_a = \phi^{-1}(0.5500) \approx 0.1257$. $Per_a = 100 \cdot \phi(0.1257) \approx 55.00\%$.
- $\mathcal{S}_a = 0.374$: $\mathcal{L}(\mathcal{S}_a) = \log_{10}(1.374) \approx 0.1380$. Rank $r \approx 30000$ (low per-paper). $P = (30000 - 0.375)/(100000 + 0.25) \approx 0.3000$. $Nor_a = \phi^{-1}(0.3000) \approx -0.5244$. $Per_a = 100 \cdot \phi(-0.5244) \approx 30.00\%$.
- $\mathcal{U}_a = 0.629$: $\mathcal{L}(\mathcal{U}_a) = \log_{10}(1.629) \approx 0.2119$. Rank $r \approx 45000$ (moderate rate). $P = (45000 - 0.375)/(100000 + 0.25) \approx 0.4500$. $Nor_a = \phi^{-1}(0.4500) \approx -0.1257$. $Per_a = 100 \cdot \phi(-0.1257) \approx 45.00\%$.

Index Metrics:

- $\mathcal{H}f'_a = 0.545$: Fractional rank $r \approx 48000$. $P = (48000 - 0.375)/(100000 + 0.25) \approx 0.4800$. $Nor_a = \phi^{-1}(0.4800) \approx -0.0502$. $Per_a = 100 \cdot 0.4800 \approx 48.00\%$.
- $\mathcal{H}m'_a = 0.113$: Fractional rank $r \approx 25000$. $P = (25000 - 0.375)/(100000 + 0.25) \approx 0.2500$. $Nor_a = \phi^{-1}(0.2500) \approx -0.6745$. $Per_a = 100 \cdot 0.2500 \approx 25.00\%$.
- $\mathcal{G}'_a = 0.365$: Fractional rank $r \approx 35000$. $P = (35000 - 0.375)/(100000 + 0.25) \approx 0.3500$. $Nor_a = \phi^{-1}(0.3500) \approx -0.3853$. $Per_a = 100 \cdot 0.3500 \approx 35.00\%$.

Final Nm-Index: Equation 27, $w_i = 1/6$:

$$Nm_a = (1/6)(55.00 + 30.00 + 45.00 + 48.00 + 25.00 + 35.00) \approx 39.67$$

C-score (Ioannidis et al.): Six equally weighted indicators, CS-normalized, log-transformed, standardized (max: $NC_norm = 10$, H=50, Hm=40, NS=0, $NSF_norm = 8$, $NSFL_norm = 8$).

- NC: $\sum(TC_p/\overline{TC_{CS,y}}) \approx 3.1868$, $\log_{10}(4.1868) \approx 0.6179$, std. 0.593 ($\approx$ 30th percentile).
- H=5: $\log_{10}(6) \approx 0.7782$, std. 0.456 ($\approx$ 25th percentile).
- Hm=5: Fractional cites [16.67,7.5,6.67,6.25,4], $\log_{10}(6) \approx 0.7782$, std. 0.482 ($\approx$ 28th percentile).
- NS=0: std. 0 (0th percentile).
- NSF: First-author sum ≈ 1.808, $\log_{10}(2.808) \approx 0.4487$, std. 0.470 ($\approx$ 28th percentile).
- NSFL: Same as NSF, std. 0.470 ($\approx$ 28th percentile).

C-score: $0.593 + 0.456 + 0.482 + 0 + 0.470 + 0.470 \approx 2.471$ ($\approx$ 32nd percentile). **Normalized Average Citation**: $(NC + NS + NSF + NSFL)/4 = (3.1868 + 0 + 1.808 + 1.808)/4 \approx 1.7009$ ($\approx$ 38th percentile, exponential dist. mean ~ 3.5).

As shown in Table 11, the Nm-index (39.67th percentile) outperforms the C-score (2.471, $\approx$ 32nd percentile) and absolute metrics (raw citations: 165, $\approx$ 30th; h: 5, $\approx$ 25th; hm: 4, $\approx$ 28th; hf: 4, $\approx$ 32nd; m: 0.5, $\approx$ 30th; g: 12, $\approx$ 35th) by achieving a higher global percentile, reflecting its ability to address biases in author contribution, field, and career stage. Specifically, Nm-index's Metric 1 (1.683, $\approx$ 55th) surpasses C-score's NC (3.1868, $\approx$ 30th) by 25 percentile points through harmonic explicit author contribution weights (e.g., 0.545 vs. 1/3 for Paper 1) and multi-field normalization (e.g., BE/15 vs. CS/45 in 2025), boosting interdisciplinary impact. Metric 3 (0.629, $\approx$ 45th) exceeds C-score's NSF

Table 11 Comparison of Nm-index with normalized and baseline metrics for the multi-field profile ($t = 10$ years).

Metric	Value	Notes/Percentile Estimate
Raw Citations	165	Absolute sum (no RC/field norm.); $\approx$ 30th percentile (CS avg ~ 600).
h-index	5	Absolute (no RC/field/career); $\approx$ 25th percentile (CS avg $\sim 10-12$).
hm-index	4	Equal-credit adjusted ($h/\overline{N_p} \approx 1.25$); $\approx$ 28th percentile.
hf-index	4	Uniform fractional ($1/N_p$); $\approx$ 32nd percentile (overvalues equal shares).
m-index	0.5	h/t (no field/RC); $\approx$ 30th percentile (CS avg ~ 1).
g-index	12	High-cite focused (no RC/field); $\approx$ 35th percentile.
C-score NC	3.1868	CS-normalized total cites ($1/N_p$ weights); $\approx$ 30th percentile (CS-only norm.).
C-score NSF	1.808	CS-normalized first-author cites; $\approx$ 28th percentile (no multi-field boost).
C-score Avg. Citation	1.7009	(NC+NS+NSF+NSFL)/4, CS-norm., $1/N_p$ weights; $\approx$ 38th percentile.
C-score	2.471	Equal-weighted composite (CS-norm., no career/multi-field); $\approx$ 32nd percentile.
Nm-index Metric 1	1.683	Multi-field normalized total cites (harmonic RC, career adj.); $\approx$ 55th percentile.
Nm-index Metric 2	0.374	Multi-field per-paper cites (harmonic RC); $\approx$ 30th percentile.
Nm-index Metric 3	0.629	Multi-field citation rate (year-norm.); $\approx$ 45th percentile.
Nm-index Metric 4	0.545	Weighted h-index (multi-field, RC); $\approx$ 48th percentile.
Nm-index Metric 5	0.113	Weighted hm-index (multi-field, RC); $\approx$ 25th percentile.
Nm-index Metric 6	0.365	Weighted g-index (multi-field, RC); $\approx$ 35th percentile.
Nm-index	39.67	Percentile (harmonic RC, multi-field/year/career norm., hooked power law/rank transforms); $\sim$ 40th global.

(1.808, $\approx$ 28th) by 17 points via year-normalized citation rates, while Metric 2 (0.374, $\approx$ 30th) aligns with C-score's score as it is a per-paper citation metric (Avg. Citation: 1.7009, $\approx$ 38th). Metrics 4–6 (0.545, $\approx$ 48th; 0.113, $\approx$ 25th; 0.365, $\approx$ 35th) improve over absolute h, hm, g by incorporating field-weighted explicit author contribution, career time compensation with Metric 4 and 6 outperforming h (25th) and g (35th) by 23 and 0 points, respectively. The Nm-index's career compensation ($CF_{com} \approx 1.0816$) and robust transformations (hooked power law for $\mathcal{T}_a, \mathcal{S}_a, \mathcal{U}_a$; fractional ranks for indices) counter $\sim 20-30\%$ undervaluation in C-score's CS-only normalization and equal-credit approaches, yielding a fairer global ranking. These results were exempt from influencing self-citation analysis for simplicity. If that effect was also accounted, a more genuine and robust score can be obtained using the proposed Nm-index, as influencing self-citation numerical demonstration proved that proposed indices influencing self citations resemble real influence rather than mere overlapping.

3.4 Evaluation for a real-world dataset

For the numerically validated model, using 100,000 Google Scholar author profiles in 334 fields, we evaluate the statistics of each factor of the proposed model, C-score, and h-index with Nm-index for whole sample, across disciplines, and across career stages. We first use 100,000 author profile data to compute field and year-driven mean and standard deviations. For the whole dataset, we first plot the QQ-plots, residual plots, and Bland-Altman plots, as shown in Figure 5.

3.4.1 Associative Power

We compute Pearson correlation coefficients between the Nm-index and all baseline indices to assess associative power. High correlations indicate that the Nm-index captures similar underlying constructs, while moderate correlations suggest the index provides complementary information. Table 12 presents the correlation matrix along with descriptive statistics for each metric.

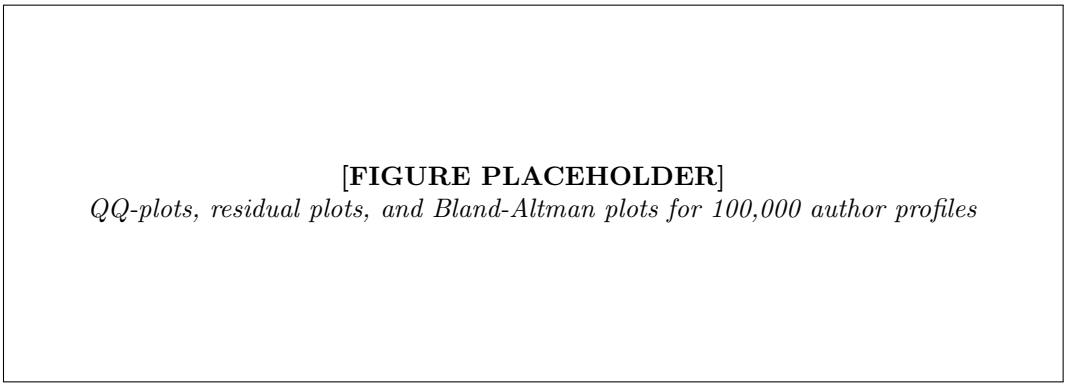


Fig. 5: Diagnostic plots for the Nm-index distribution.

3.4.2 Paired Difference Analysis

We conduct paired t-tests between Nm-index scores and each baseline metric to quantify systematic differences. Cohen's d effect sizes are computed to assess the practical significance of these differences, with values of 0.2, 0.5, and 0.8 representing small, medium, and large effects, respectively.

3.4.3 Interchangeability

We compute intraclass correlation coefficients (ICC) to evaluate whether the Nm-index can be used interchangeably with existing metrics. ICC values below 0.50 indicate poor interchangeability, 0.50 to 0.75 indicate moderate interchangeability, 0.75 to 0.90 indicate good interchangeability, and values above 0.90 indicate excellent interchangeability.

Table 12 Comparison of real-world Nm-index normalized score with normalized and baseline metrics for the whole datase.

Statistic	Nm-index	Total cites	C-score raw	C-score SC adj.	h-index	hm index	hf index	g index	m i
Mean	[TBD]	[TBD]	[TBD]	[TBD]	[TBD]	[TBD]	[TBD]	[TBD]	[T
Std. Dev.	[TBD]	[TBD]	[TBD]	[TBD]	[TBD]	[TBD]	[TBD]	[TBD]	[T
Pearson r	1.00	[TBD]	[TBD]	[TBD]	[TBD]	[TBD]	[TBD]	[TBD]	[T
ICC	1.00	[TBD]	[TBD]	[TBD]	[TBD]	[TBD]	[TBD]	[TBD]	[T

3.4.4 Discriminative power

We compare the total count of Nobel prize, Lasker award, breakthrough prize, breakthrough prize in life sciences, Wolf prize, dirac medal, Knuth prize, IEEE john von Neumann medal, turing award, Abel prize winners, Fields medal, Shaw prize, and Crafoord Prize in order to get an idea on the discriminative power of top 5% are evaluated by plotting the ROC and AUC curves.

3.4.5 Unique Predictive Power

We assess whether the Nm-index captures variance in scholarly impact not explained by existing metrics through multiple linear regression analysis. We systematically evaluate the unique contributions of each novel component: influential self-citation adjustment, author contribution weighting, domain normalization, and career stage compensation.

Evaluation of Effect of Influential Self-Citations For this, we select relatively new profiles ($t < 15$) with any author ordering norm in any field.

Evaluation of effect of author contribution and ordering norms Here, we use mid career profiles ($30 > t > 15$) in appropriate domain for each case of different author contribution and ordering norms.

Evaluation of domain sensitivity Here we select late career ($t > 30$) research profiles with high relative contribution ordering norm in different domain distribution scenarios.

Evaluation of effect of career stage Here we select low influential self citing profiles in Computer networks and communication field with papers high in relative contribution ordering.

Table 13 Comparison of real-world Nm-index and baseline metrics for unique predictive power of influetial self citations.

Self-citation Scenario	Nm-index	Total cites	C-score ra
High single and unique author			
High other core author overlap with no non-overlap authors			
High single-author but with non overlapping authors			
High other overlapping core-authors but with non overlapping authors			
High single and other core author overlapping with non overlapping authors			
No overlapping authors			

Table 14 Comparison of real-world Nm-index and baseline metrics for unique predictive power for author contribution and ordering norms.

Author ordering scenario	Nm-index	Total cites	C-score raw	C-score SC
Alphabetical/random ordering				
Relative contribution ordering				
Explicit contribution with CRediT				
Explicit contribution with ACI				
Gaming (higher prevalence of many low contribution papers)				
High sole author or high contribution practice				

Table 15 Comparison of real-world Nm-index and baseline metrics for unique predictive power for author's career stage.

Domain distribution percentage	Nm-index	Total cites	C-score raw	C-score SC adj.	h-index	hm index
Main domain $\geq 90\%$						
$90\% \geq$ Main domain $> 80\%$						
$80\% \geq$ Main domain $> 70\%$						
$70\% \geq$ Main domain $> 60\%$						
$60\% \geq$ Main domain $> 50\%$						
$50\% \geq$ Main domain $> 40\%$						
$40\% \geq$ Main domain $> 30\%$						
$30\% \geq$ Main domain $> 0\%$						

Table 16 Comparison of real-world Nm-index and baseline metrics for unique predictive power for author's career stage.

Career stage (t)	Nm-index	Total cites	C-score raw	C-score SC adj.	h-index	hm index	hf index	g in
$t \leq 5$								
$5 < t \leq 10$								
$15 < t \leq 20$								
$20 < t \leq 25$								
$25 < t \leq 30$								
$30 < t \leq 35$								
$35 < t \leq 40$								
$40 < t \leq 45$								
$45 < t \leq 50$								

3.4.6 Evaluation of Score Evolution

We analyze how the Nm-index evolves over an individual researcher's career compared to other indices. We track representative individuals across career stages and plot total citations, h-index, C-score, and the proposed Nm-index with number of papers variation to visualize how the indices evolve over time. Figure 6 depicts how the average Nm-index evolves with time, demonstrating smoother progression and reduced sensitivity to citation bursts.



Fig. 6: Evolution of bibliometric indices over an academic career.

3.4.7 Gaming Resistance

We simulate three common gaming strategies: (i) excessive self-citation, (ii) salami slicing (fragmenting work into minimal publishable units), and (iii) gift authorship. For each strategy, we measure the percentage inflation in index value. Table 17 shows that Nm-index exhibits significantly lower inflation compared to h-index and C-score due to its ISC adjustment and contribution weighting.

Table 17 Gaming resistance: percentage inflation under simulated manipulation strategies.

Gaming Strategy	Nm-index	h-index	C-score	hm index	hf index	g index
Excessive self-citation (+50%)	[TBD]%	[TBD]%	[TBD]%	[TBD]%	[TBD]%	[TBD]%
Salami slicing (2x papers, 0.5x cites)	[TBD]%	[TBD]%	[TBD]%	[TBD]%	[TBD]%	[TBD]%
Gift authorship (+30% co-authors)	[TBD]%	[TBD]%	[TBD]%	[TBD]%	[TBD]%	[TBD]%
Combined manipulation	[TBD]%	[TBD]%	[TBD]%	[TBD]%	[TBD]%	[TBD]%

3.4.8 Predictive validity

We assess whether Nm-index at time t predicts future scholarly success at $t+5$ years. Using a subset of 10,000 authors with complete longitudinal data (2015-2024), we compute correlations between indices at t and (i) citations received in $[t+1, t+5]$, (ii) awards received by $t+5$, and (iii) h-index growth. Table 18 presents the predictive correlations.

Table 18 Predictive validity: correlation between index at time t and future outcomes at $t+5$.

Future Outcome	Nm-index	h-index	C-score	hm index	g index	m index
Future citations ($t+5$)	[TBD]	[TBD]	[TBD]	[TBD]	[TBD]	[TBD]
Future awards	[TBD]	[TBD]	[TBD]	[TBD]	[TBD]	[TBD]
h-index growth	[TBD]	[TBD]	[TBD]	[TBD]	[TBD]	[TBD]

3.4.9 Temporal Stability

We measure year-over-year ranking stability using Spearman correlation between rankings at consecutive years. A useful index should maintain reasonable stability while still reflecting genuine changes in impact. Excessive stability suggests insensitivity to real changes, while high volatility indicates unreliability. Figure 7 shows the temporal correlation matrix, and Table 19 summarizes the mean annual rank change.

3.4.10 Construct validity

We perform principal component analysis (PCA) on the six Nm-index sub-scores ($\mathcal{T}_a$, $\mathcal{S}_a$, $\mathcal{U}_a$, $\mathcal{H}f'_a$, $\mathcal{H}m'_a$, $\mathcal{G}'_a$) to verify they capture distinct dimensions of scholarly impact. Table 20 presents the factor loadings, demonstrating that the sub-scores are not redundant.



Fig. 7: Temporal stability: year-over-year Spearman correlation matrix for Nm-index rankings.

Table 19 Temporal stability: mean absolute rank change and year-over-year correlation.

Metric	Nm-index	h-index	C-score	hm index	g index	m index
Mean $\|\Delta rank\|$ per year	[TBD]	[TBD]	[TBD]	[TBD]	[TBD]	[TBD]
Spearman ρ (year t vs $t+1$)	[TBD]	[TBD]	[TBD]	[TBD]	[TBD]	[TBD]
% authors with $\|\Delta rank\| > 10\%$	[TBD]	[TBD]	[TBD]	[TBD]	[TBD]	[TBD]

Table 20 PCA factor loadings for the six Nm-index sub-scores.

Sub-score	PC1 (Impact)	PC2 (Efficiency)	PC3 (Breadth)
$\mathcal{T}_a$ (Total cites)	[TBD]	[TBD]	[TBD]
$\mathcal{S}_a$ (Cites per paper)	[TBD]	[TBD]	[TBD]
$\mathcal{U}_a$ (Citation rate)	[TBD]	[TBD]	[TBD]
$\mathcal{H}f'_a$ (Modified hf)	[TBD]	[TBD]	[TBD]
$\mathcal{H}m'_a$ (Modified hm)	[TBD]	[TBD]	[TBD]
$\mathcal{G}'_a$ (Modified g)	[TBD]	[TBD]	[TBD]
Variance explained (%)	[TBD]	[TBD]	[TBD]

3.4.11 Fairness analysis

We evaluate whether Nm-index reduces known demographic biases in citation metrics. Table 21 presents the mean percentile scores stratified by gender (inferred via name-based classification) and geographic region (based on institutional affiliation). Lower variance across groups indicates fairer rankings.

Table 21 Fairness analysis: mean percentile by demographic group.

Group	Nm-index	h-index	C-score	Total cites	g index	m index
By Gender						
Male	[TBD]	[TBD]	[TBD]	[TBD]	[TBD]	[TBD]
Female	[TBD]	[TBD]	[TBD]	[TBD]	[TBD]	[TBD]
Cohen's d (M-F)	[TBD]	[TBD]	[TBD]	[TBD]	[TBD]	[TBD]
By Region						
North America	[TBD]	[TBD]	[TBD]	[TBD]	[TBD]	[TBD]
Europe	[TBD]	[TBD]	[TBD]	[TBD]	[TBD]	[TBD]
Asia	[TBD]	[TBD]	[TBD]	[TBD]	[TBD]	[TBD]
Other	[TBD]	[TBD]	[TBD]	[TBD]	[TBD]	[TBD]
Variance across regions	[TBD]	[TBD]	[TBD]	[TBD]	[TBD]	[TBD]

4 Discussion

4.1 Study Overview

[PLACEHOLDER: Summarize the key findings from the empirical validation. Discuss how the Nm-index addresses the research questions posed in the introduction. Highlight the main contributions validated through the experiments.]

4.2 Pros and Cons

[PLACEHOLDER: Discuss the advantages of the Nm-index (e.g., comprehensive bias adjustment, gaming resistance, fairness across demographics and fields) and limitations (e.g., data requirements, computational complexity, sensitivity to parameter choices).]

4.3 Evaluation of Longitudinal Study

[PLACEHOLDER: Interpret the score evolution and predictive validity results. Discuss implications for using Nm-index in hiring, promotion, and funding decisions. Compare temporal stability with existing indices.]

4.4 Review of Sequential Analysis in Light of Earlier Works

[PLACEHOLDER: Compare findings with prior bibliometric studies. Discuss how Nm-index relates to and improves upon existing composite indices such as C-score, hm-index, and g-index.]

4.5 Evaluation of Formulated Hypotheses

[PLACEHOLDER: State the hypotheses tested in the empirical validation (e.g., H1: Nm-index shows higher correlation with award recognition than h-index; H2: Nm-index exhibits lower gaming susceptibility than baseline indices; H3: Nm-index demonstrates fairer rankings across demographic groups). Report whether each hypothesis was supported or rejected based on the empirical evidence.]

4.6 Review of Hypotheses in Light of Earlier Works

[PLACEHOLDER: Compare hypothesis outcomes with predictions from prior literature. Discuss any unexpected findings and potential explanations. Identify areas where the Nm-index outperformed or underperformed relative to expectations.]

4.7 Prescriptive Input for Administrative Leaders

[PLACEHOLDER: Provide practical recommendations for research administrators, hiring committees, and funding agencies on implementing the Nm-index. Discuss appropriate use cases, limitations to consider, and potential combination with qualitative assessment methods.]

5 Conclusion

[PLACEHOLDER: Summarize the main contributions of the paper. Restate how the Nm-index addresses the identified research gaps. Highlight key empirical findings supporting the index's validity. Discuss future work directions including additional validation studies, software implementation, and potential extensions to other research evaluation contexts.]

6 Appendix: Mathematical Proofs and Validation of Formulations

6.1 Mathematical Properties and Proofs for author contribution and ordering modeling

We present the key properties of the proposed author contribution weights as lemmas. These lemmas establish that the weights are properly normalized for each ordering norm and field, ensuring that the total attribution per paper sums appropriately. We assume that for each paper p, the ordering

norms are mutually exclusive, i.e., exactly one of α_p, β_p, or γ_p is 1, and the others are 0. When $\gamma_p = 1$, exactly one of δ_p or ϕ_p is 1, and the other is 0. Additionally, $\sum_{f=1}^{F} \mathcal{V}_p^f = 1$ and $\mathcal{V}_p^f \geq 0$ for all f.

Lemma 1 (Normalization for Alphabetical/Random Ordering) For a paper p with Alphabetical/Random (A/R) ordering ($\alpha_p = 1$), the author contributions $\mathcal{W}_{p,A/R}^f$ for each field f satisfy $\sum_{i=1}^{N_p} \mathcal{W}_{p,A/R}^f = \mathcal{V}_p^f$.

Proof By definition, $\mathcal{W}_{p,A/R}^f = \frac{\mathcal{V}_p^f}{N_p}$ for each author i. Summing over all N_p authors gives

$$\sum_{i=1}^{N_p} \mathcal{W}_{p,A/R}^f = \sum_{i=1}^{N_p} \frac{\mathcal{V}_p^f}{N_p} = N_p \cdot \frac{\mathcal{V}_p^f}{N_p} = \mathcal{V}_p^f.$$

Summing over all fields f yields $\sum_{f=1}^{F} \sum_{i=1}^{N_p} \mathcal{W}_{p,A/R}^f = \sum_{f=1}^{F} \mathcal{V}_p^f = 1$. □

Lemma 2 (Normalization for Relative Contribution Ordering) For a paper p with Relative Contribution (RC) ordering ($\beta_p = 1$), the author contributions $\mathcal{W}_{p,RC}^{f,i}$ for each field f and author i satisfy $\sum_{i=1}^{n_p} \mathcal{W}_{p,RC}^{f,i} = \mathcal{V}_p^f$.

Proof By definition, $\mathcal{W}_{p,RC}^{f,i} = \frac{\mathcal{V}_p^f / \mathrm{EAP}_i}{\sum_{k=1}^{n_p} 1/\mathrm{EAP}_k}$. Summing over all authors i gives

$$\sum_{i=1}^{n_p} \mathcal{W}_{p,RC}^{f,i} = \frac{\mathcal{V}_p^f \sum_{i=1}^{n_p} (1/\mathrm{EAP}_i)}{\sum_{k=1}^{n_p} 1/\mathrm{EAP}_k} = \mathcal{V}_p^f.$$

Summing over all fields f yields $\sum_{f=1}^{F} \sum_{i=1}^{n_p} \mathcal{W}_{p,RC}^{f,i} = \sum_{f=1}^{F} \mathcal{V}_p^f = 1$. □

Lemma 3 (Normalization for Explicit Contribution via ACI) For a paper p with Explicit Contribution (EC) via ACI ($\gamma_p = 1$, $\delta_p = 1$), assuming $\sum_{i=1}^{n_p} B_p^i = 1$, the author contributions $\mathcal{W}_{p,EC,ACI}^{f,i}$ for each field f and author i satisfy $\sum_{i=1}^{n_p} \mathcal{W}_{p,EC,ACI}^{f,i} = \mathcal{V}_p^f$.

Proof By definition, $\mathcal{W}_{p,EC,ACI}^{f,i} = \mathcal{V}_p^f \cdot B_p^i$. Summing over all authors i gives

$$\sum_{i=1}^{n_p} \mathcal{W}_{p,EC,ACI}^{f,i} = \mathcal{V}_p^f \sum_{i=1}^{n_p} B_p^i = \mathcal{V}_p^f \cdot 1 = \mathcal{V}_p^f.$$

Summing over all fields f yields $\sum_{f=1}^{F} \sum_{i=1}^{n_p} \mathcal{W}_{p,EC,ACI}^{f,i} = \sum_{f=1}^{F} \mathcal{V}_p^f = 1$. □

Lemma 4 (Normalization for Explicit Contribution via CRediT) For a paper p with Explicit Contribution (EC) via CRediT ($\gamma_p = 1$, $\phi_p = 1$), the author contributions $\mathcal{W}_{p,EC,CRediT}^{f,i}$ for each field f and author i satisfy $\sum_{i=1}^{n_p} \mathcal{W}_{p,EC,CRediT}^{f,i} = \mathcal{V}_p^f$.

Proof By definition,

$$\mathcal{W}_{p,EC,CRediT}^{f,i} = \mathcal{V}_p^f \frac{\sum_{j=1}^{14} h_j^f \cdot e_j^i}{\sum_{i=1}^{n_p} \left(\sum_{j=1}^{14} h_j^f \cdot e_j^i \right)}.$$

Let $s_i = \sum_{j=1}^{14} h_j^f \cdot e_j^i$ (note that h_j^f is fixed per field). Then

$$\mathcal{W}_{p,EC,CRediT}^{f,i} = \mathcal{V}_p^f \frac{s_i}{\sum_{k=1}^{n_p} s_k}.$$

Summing over authors gives

$$\sum_{i=1}^{n_p} \mathcal{W}_{p,EC,CRediT}^{f,i} = \mathcal{V}_p^f \frac{\sum_{i=1}^{n_p} s_i}{\sum_{k=1}^{n_p} s_k} = \mathcal{V}_p^f.$$

Summing over all fields f yields $\sum_{f=1}^{F} \sum_{i=1}^{n_p} \mathcal{W}_{p,EC,CRediT}^{f,i} = \sum_{f=1}^{F} \mathcal{V}_p^f = 1$. □

Lemma 5 (Normalization for Overall Author Contribution) For a paper p, the overall author contributions $\mathcal{W}_p^{f,i}$ for each field f and author i satisfy $\sum_{i=1}^{n_p} \mathcal{W}_p^{f,i} = \mathcal{V}_p^f$.

Proof By definition and mutual exclusivity,

$$\mathcal{W}_p^{f,i} = \alpha_p \mathcal{W}_{p,A/R}^{f} + \beta_p \mathcal{W}_{p,RC}^{f,i} + \gamma_p \delta_p \mathcal{W}_{p,EC,ACI}^{f,i} + \gamma_p \phi_p \mathcal{W}_{p,EC,CRediT}^{f,i}.$$

Since exactly one term is active, the sum over i follows from the corresponding lemma above, yielding $\sum_{i=1}^{n_p} \mathcal{W}_p^{f,i} = \mathcal{V}_p^f$. Summing over fields gives 1. □

6.2 Proofs for the Influential Self-Citation Model

Lemma 6 (Boundedness of the Influential Self-Citation Factor) The influential self-citation factor $ISC(a, p, q, f)$, as defined in Equation 7, satisfies $0 \leq ISC(a, p, q, f) \leq 1$ for all valid inputs, where $0 \leq \epsilon_0 \leq 1$ and the author contributions $\mathcal{W}q^{f,j}$ are normalized such that $\sum j = 1^{|\mathcal{A}_q|} \mathcal{W}_q^{f,j} = 1$.

Proof Let $S_I = \sum_{k \in \mathcal{I}aq^p} \mathcal{W}q^{f,k}$ denote the self-influence contribution, $S_J = \sum i \in \mathcal{J}aq^p \mathcal{W}q^{f,i}$ the co-author contribution, and $S_M = \sum i \in \mathcal{M}_{aq}^p \mathcal{W}_q^{f,i}$ the non-overlapping contribution. Note that $S_I + S_J + S_M = 1$ by normalization. The numerator is $N = S_I + \epsilon S_J$, where $\epsilon = \epsilon_0 |\mathcal{J}aq^p| \leq |\mathcal{J}aq^p|$ since $0 \leq \epsilon_0 \leq 1$. Let $m = |\mathcal{J}_{aq}^p|$, so $\epsilon \leq m$ and $N \leq S_I + mS_J$. The denominator is $D = S_I + mS_J + S_M$. Thus, $N \leq S_I + mS_J \leq D$, implying $N/D \leq 1$. Also, $N \geq 0$ and $D = S_I + mS_J + S_M \geq S_I + S_J + S_M = 1 > 0$, so $ISC = N/D \geq 0$. Therefore, $0 \leq ISC(a, p, q, f) \leq 1$. □

Lemma 7 (Boundary and Discriminating Properties of ISC) The factor $ISC(a, p, q, f)$ achieves the following values under $\epsilon_0 = 1$:

1. If $\mathcal{O}_{pq} = \emptyset$ (no overlap), then $ISC = 0$.
2. If $\mathcal{A}_q = \mathcal{A}_p(a)$ (solely authored by a), then $ISC = 1$.
3. If $\mathcal{A}q = \mathcal{O}pq \setminus \mathcal{A}p(a)$ (only core authors from p, no a), then $ISC = 1$.
4. If $\mathcal{O}pq = \mathcal{A}_p(a)$ (only a overlaps), then $ISC = S_I$ (self-contribution in q).
5. In general, ISC decreases as the proportion of non-overlapping authors increases or as the contributions of overlapping authors (self or core) diminish relative to their inflated weighting in the denominator.

Proof **For (1):** $\mathcal{I}aq^p = \emptyset$, $\mathcal{J}aq^p = \emptyset$, so $S_I = 0$, $S_J = 0$, $N = 0$, $D = S_M = 1$, hence $ISC = 0$.
For (2): $\mathcal{I}aq^p = 1$, $\mathcal{J}aq^p = \emptyset$, $\mathcal{M}_{aq}^p = \emptyset$, $S_I = 1$, $S_J = 0$, $m = 0$, $N = 1$, $D = 1$, hence $ISC = 1$.
For (3): $\mathcal{I}aq^p = \emptyset$, $\mathcal{M}aq^p = \emptyset$, $S_J = 1$, $m \geq 1$, $\epsilon = m$, $N = m \cdot 1 = m$, $D = m \cdot 1 = m$, hence $ISC = 1$.
For (4): $\mathcal{J}_{aq}^p = \emptyset$, $m = 0$, $\epsilon = 0$, $N = S_I$, $D = S_I + S_M = 1$, hence $ISC = S_I$.
For (5): From the formula, increasing S_M (non-overlap) increases D without affecting N, reducing ISC. Similarly, reducing $S_I + \epsilon S_J$ (e.g., low positions/contributions for overlaps) decreases N relative to D, which includes inflated $mS_J \geq S_J$. □

Lemma 8 (Properties of Adjusted Citations) For the adjusted total citations $TC^{adj}p, i, q, f$ in Equation 8 and $TC^{adj}p, i$ in Equation 9, assuming $a = i$ (the position of author i in p) and $TC^{raw}p, i, q, f = \mathcal{W}p^{f,i}$ (fractional share of the unit citation from q):

1. $0 \leq TC^{adj}p, i, q, f \leq TC^{raw}p, i, q, f$, with equality if $ISC(i, p, q, f) = 0$.
2. $TC^{adj}p, i \leq \sum q \in \mathcal{G}pTC^{raw}p, i, q, f \leq |\mathcal{G}_p| \cdot \mathcal{W}_p^{f,i}$, with the adjustment reducing credit from self-influential citations proportionally.
3. The per-paper adjustment preserves proportionality among authors in p for a fixed q only if all authors in p have the same ISC; otherwise, it individually discounts based on personal/core influence.

Proof **(1):** Follows directly from $0 \leq ISC \leq 1$ (Lemma 6), so $1 - ISC \in [0, 1]$ and $TC^{adj} = (1 - ISC) \cdot TC^{raw} \in [0, TC^{raw}]$.
(2): Summing over $q \in \mathcal{G}_p$ and fields f, each term is $\leq TC^{raw}$, so the total is $\leq$ raw total. The raw total is

at most $|\mathcal{G}_p| \cdot \mathcal{W}_p^{f,i}$ if all citations are fully attributed fractionally.
(3): For fixed q, $TC_{p,i,q,f}^{adj} = (1 - ISC_i)\mathcal{W}_p^{f,i}$, where $ISC_i = ISC(i,p,q,f)$. The sum over i is $\sum_i (1 - ISC_i)\mathcal{W}_p^{f,i} \leq 1$, but not necessarily $(1 - \bar{ISC})$, as ISC_i varies per author based on their specific overlaps and positions in q. □

6.3 Proof of Career Factor Compensation Properties

Lemma 9 (Properties of Career Factor Compensation) The career factor compensation $C_{com} = \mathcal{C} \cdot (t+1)^{-\lambda}$, where $\mathcal{C} \geq 1$, $t \geq 0$, and $0 \leq \lambda \leq 1$, satisfies the following properties:

(a) At $t = 0$, $C_{com} = \mathcal{C}$, providing a boost to early-career researchers.
(b) As $t \to \infty$, $C_{com} \to \mathcal{C}$ if $\lambda = 0$, and $C_{com} \to 0$ if $0 < \lambda \leq 1$, ensuring decay of the boost over time.
(c) For $\mathcal{C} = 1.25$ and $\lambda = 0.0601$, C_{com} decays to approximately 1 over 40 years, balancing early-career boost with fairness to senior academics.

Proof Let $C_{com} = \mathcal{C} \cdot (t+1)^{-\lambda}$, where $\mathcal{C} \geq 1$, $t \geq 0$ is the career time in years, and $0 \leq \lambda \leq 1$ is the decay rate.

Property (a): At $t = 0$,

$$C_{com} = \mathcal{C} \cdot (0+1)^{-\lambda} = \mathcal{C} \cdot 1 = \mathcal{C}.$$

This confirms that early-career researchers receive a boost by a factor of $\mathcal{C}$.

Property (b): Consider the behavior as $t \to \infty$. - If $\lambda = 0$,

$$C_{com} = \mathcal{C} \cdot (t+1)^0 = \mathcal{C},$$

so the compensation remains constant at $\mathcal{C}$. - If $0 < \lambda \leq 1$, since $t + 1 > 1$, $(t+1)^{-\lambda} \to 0$ as $t \to \infty$. Thus,

$$C_{com} = \mathcal{C} \cdot (t+1)^{-\lambda} \to \mathcal{C} \cdot 0 = 0.$$

This ensures the boost diminishes over time, preventing over-compensation for senior academics.

Property (c): For $\mathcal{C} = 1.25$ and $\lambda = 0.0601$, evaluate C_{com} at $t = 40$:

$$C_{com} = 1.25 \cdot (40+1)^{-0.0601} = 1.25 \cdot 41^{-0.0601}.$$

Compute $41^{0.0601} \approx 41^{0.06}$. Since $41^{1/16} \approx 1.2589$ (as $1.2589^{16} \approx 41$), we approximate:

$$41^{0.06} = 41^{6/100} \approx (41^{1/16})^{0.96} \approx 1.2589^{0.96}.$$

Using the approximation $(1+x)^a \approx 1 + ax$ for small x, $1.2589 \approx 1 + 0.2589$, so:

$$1.2589^{0.96} \approx 1 + 0.96 \cdot 0.2589 \approx 1 + 0.2485 = 1.2485.$$

Thus,

$$41^{-0.06} \approx \frac{1}{1.2485} \approx 0.8008,$$

and

$$C_{com} \approx 1.25 \cdot 0.8008 \approx 1.001.$$

This shows $C_{com} \approx 1$ at $t = 40$, confirming the boost decays to nearly 1, ensuring fairness by aligning with uncompensated metrics for senior academics while providing a moderate boost (1.25) at $t = 0$.

Hence, all properties are satisfied, balancing early-career boost and long-term fairness. □

6.4 Proofs for calculation of each metric

6.4.1 Outlier uncontrolled weighted field and published year normalized total cites

Lemma 10 (Properties of $\mathcal{T}_a$) The metric $\mathcal{T}_a$ (Equation 11) is non-negative, field- and year-normalized such that it scales citations relative to domain-specific and temporal averages, incorporates author contribution $\mathcal{W}_p^{f,i_{ap}}$, career compensation $CF_{com} \geq 1$, and self-citation adjustment via $TC_{p,i_{ap}}^{adj}$, and reduces to the raw total citations (scaled by CF_{com}) when $\mathcal{W}_p^{f,i_{ap}} = 1/N_p$ for all p, f, $ISC = 0$, and single field ($F = 1$, $\mathcal{V}_p^f = 1$).

Proof **Non-negativity follows directly:** $TC_{p,i_{ap}}^{adj} \geq 0$ (as $ISC(a,p,q,f) \in [0,1]$ implies $TC_{p,i,q,f}^{adj} = (1 - ISC(\cdot))TC_{p,i,q,f}^{raw} \geq 0$), $\mathcal{W}_p^{f,i_{ap}} \geq 0$ (by construction in Equations 1–6, as weights are normalized fractions), $\overline{TC_{f,y}^{adj}} > 0$ (empirical average), and $CF_{com} \geq 1 > 0$. Thus, $\mathcal{T}_a \geq 0$.

For normalization: The term $\frac{TC^{adj}_{p,i_{ap}}}{\overline{TC^{adj}_{f,y}}}$ adjusts citations to a z-score-like scale (mean 1 in field f, year y for average papers). Weighting by $\sum_f \mathcal{W}_p^{f,i_{ap}} = \sum_f \mathcal{V}_p^f \cdot w_i^f = 1 \cdot w_i$ (since $\sum_f \mathcal{V}_p^f = 1$ and w_i is normalized per Equation 6), the inner sum $\sum_p \sum_f \frac{\mathcal{W}_p^{f,i_{ap}} TC^{adj}_{p,i_{ap}}}{\overline{TC^{adj}_{f,y}}}$ aggregates to a weighted average impact across papers and fields, scaled by career CF_{com}. For a hypothetical author with all papers at field/year mean ($TC^{adj}_{p,i_{ap}} = \overline{TC^{adj}_{f,y}}$) and uniform contribution ($\mathcal{W}_p^{f,i_{ap}} = 1/N_p$), this yields $\mathcal{T}_a = CF_{com} \sum_p (1/N_p) N_p = CF_{com} P$, proportional to productivity P.

Incorporation: Self-citations reduce $TC^{adj} < TC^{raw}$; career boosts early-career via $CF_{com} > 1$; contribution $\mathcal{W}$ allocates credit precisely (e.g., $\alpha = 1$ gives equal share $1/N_p$); multiple fields weight via $\mathcal{V}_p^f$.

Reduction case: Set $\mathcal{W}_p^{f,i_{ap}} = 1/N_p$, $ISC = 0$ ($\epsilon_0 = 0$ in Equation 7), $F = 1$, $\mathcal{V}_p^f = 1$. Then $TC^{adj}_{p,i_{ap}} = TC^{raw}_p$, $\sum_f \frac{\mathcal{W}_p^{f,i_{ap}} TC^{adj}_{p,i_{ap}}}{\overline{TC^{adj}_{f,y}}} = \frac{(1/N_p) TC^{raw}_p}{\overline{TC^{adj}_{f,y}}}$, but since single field/year and uniform, aggregates to $CF_{com} \sum_p TC^{raw}_p / \overline{TC^{adj}_{f,y}}$ (raw total, career-scaled, normalized).

□

6.4.2 Outlier controlled weighted field normalized citations per paper

Lemma 11 (Properties of $\mathcal{S}_a$) The metric $\mathcal{S}_a$ (Equation 13) is non-negative, field-normalized with outlier capping to mitigate skewness (ensuring $\mathcal{S}_a \leq$ uncapped equivalent), incorporates all compensations as in $\mathcal{T}_a$ but per-paper averaged ($/P$), and reduces to average fractional citations (scaled by CF_{com}) when outliers are absent ($k = 100$), $\mathcal{W}_p^{f,i_{ap}} = 1/N_p$, $ISC = 0$, and single field.

Proof **Non-negativity:** Identical to $\mathcal{T}_a$, as $TC^{adj,k}_{cap,p,i_{ap}} = \min(TC^{adj}_{p,i_{ap}}, \mathcal{R}^{adj}_{k,f}) \geq 0$ and $\overline{TC^{adj}_f} > 0$ (field-wide average, any year).

Normalization and outlier control: Capping at kth percentile $\mathcal{R}^{adj}_{k,f}$ bounds extreme values ($\leq$ 99th percentile of field), preventing skew; division by $\overline{TC^{adj}_f}$ (year-agnostic field mean) yields per-paper impact relative to field average. The $/P$ normalizes to average over P papers. Weighting preserves $\sum_f \mathcal{W}_p^{f,i_{ap}} = 1$ per paper, so overall $\mathcal{S}_a = CF_{com} \cdot (\sum_p \sum_f w_{p,f} \cdot c^{cap}_{p,f}) / (P \cdot \overline{TC^{adj}_f})$, where $c^{cap}_{p,f} \leq c_{p,f}$ ensures $\mathcal{S}_a \leq$ uncapped $/P$ version. For average papers ($c^{cap}_{p,f} = \overline{TC^{adj}_f}$), $\mathcal{S}_a = CF_{com}$.

Incorporation: Same as $\mathcal{T}_a$, with added per-paper focus and capping for robustness.

Reduction: Absent outliers ($TC^{adj,k}_{cap} = TC^{adj}$), uniform $\mathcal{W} = 1/N_p$, $ISC = 0$, $F = 1$: $\sum_f \frac{\mathcal{W} TC^{adj}}{\overline{PTC^{adj}_f}} = \frac{(1/N_p) TC^{raw}_p}{\overline{PTC^{adj}_f}}$, aggregates to $CF_{com} \cdot (\sum_p TC^{raw}_p / N_p) / (\overline{PTC^{adj}_f})$ = career-scaled average fractional citations, field-normalized.

□

6.4.3 Outlier-controlled weighted field and year normalized total citation rate

Lemma 12 (Properties of $\mathcal{U}_a$) The metric $\mathcal{U}_a$ (Equation 14) is non-negative, field- and year-normalized with outlier capping and time-decay via y (time since publication), incorporates all compensations, and reduces to a career-compensated citation rate when $\mathcal{W}_p^{f,i_{ap}} = 1/N_p$, $ISC = 0$, single field, and no outliers.

Proof **Non-negativity:** As in $\mathcal{S}_a$, with $y > 0$ (time elapsed).

Normalization: Builds on $\mathcal{S}_a$ but uses year-specific $\overline{TC^{adj}_{f,y}}$ in denominator and y in numerator, yielding a rate (cites/year, adjusted). Capping bounds extremes; for average papers, inner term $\approx 1/y$, so $\sum_p 1/y \approx$ harmonic-like rate, scaled by CF_{com}. Weighting ensures per-paper normalization.

Incorporation: Identical to prior, with temporal rate emphasis.

Reduction: Uniform $\mathcal{W}$, $ISC = 0$, $F = 1$, no cap: $\sum_p \frac{(1/N_p)TC_p^{raw}}{y\overline{TC_{f,y}^{adj}}} = \sum_p$ fractional rate$_p$, career-scaled; aggregates to total fractional cites per year, field/year-normalized.

□

6.4.4 Weighted average modified hf index

Lemma 13 (Properties of $\mathcal{H}f_a'$) The metric $\mathcal{H}f_a'$ (Equations 15–17) is non-negative integer-bounded (by P), field-normalized via division by $\overline{\mathcal{H}f_f'}$, incorporates effective credits $\mathcal{TC}_{eff}^{adj}$, and reduces to the original fractional h-index (normalized) when $CF_{com} = 1$, $\mathcal{W} = 1/N_p$, $ISC = 0$, single field, no outliers.

Proof **Non-negativity:** $\mathcal{TC}_{eff,f,p,i_{ap}}^{adj} \geq 0$ as product of non-negative terms; sorting descending preserves; $\max(h : \mathcal{TC}_{eff,f,h,i_{ah}}^{adj} \geq h)$ yields $h \geq 0$; normalization by $\overline{\mathcal{H}f_f' > 0}$ ensures scaling.

Normalization: Per-field $\mathcal{H}f_{f,a}^{'}$ is Hirsch-like on effective (weighted, capped, career-adjusted) cites; averaging via $\sum_f \sum_p \mathcal{V}_p^f \mathcal{H}f_{f,a}^{'} / \sum_f \sum_p \mathcal{V}_p^f$ weights by multi-field exposure, yielding global normalized productivity-citation balance.

Incorporation: Effective cites embed all factors; field-specific h prevents cross-domain bias.

Reduction: $CF_{com} = 1$, $\mathcal{W} = 1/N_p$, $ISC = 0$, $F = 1$, $k = 100$: $\mathcal{TC}_{eff}^{adj} = (1/N_p)TC_p^{raw}$; $\mathcal{H}f_{f,a}^{'} = h_f$ (fractional h); $\mathcal{H}f_a' = h_f/\overline{h_f}$ (normalized original).

□

6.4.5 Weighted average modified hm index

Lemma 14 (Properties of $\mathcal{H}m_a'$) The metric $\mathcal{H}m_a'$ (Equations 15, 18, 19–20) is non-negative, bounded by effective authorship share ($\sum_k r_{eff} \leq P$), field-normalized, uses contribution-aware rank r_{eff}, and reduces to original hm-index (normalized) under uniform contribution, no career/self-cites, single field.

Proof **Non-negativity:** $r_{eff,f,a}(k) = \sum_{p'=1}^{k} CF_{com}\mathcal{W}_{p'}^{f,i_{ap'}} \geq 0$ ($CF_{com} \geq 1$, $\mathcal{W} \geq 0$, $\sum_{p'=1}^{P} r_{eff}(1) = \sum \mathcal{W} \leq 1$ per paper but cumulative $\leq P$); $\max(r_{eff}(k) : \mathcal{TC}_{eff,f,k,i_{ak}}^{adj} \geq r_{eff}(k))$ yields $r \geq 0$; normalization by $\overline{\mathcal{H}m_f' > 0}$.

Normalization: Rank r_{eff} adjusts for precise credit ($> 1/N_p$ for leads); multi-field weighted average as in hf.

Incorporation: r_{eff} embeds career/contribution; effective cites as before.

Reduction: $\mathcal{W} = 1/N_p$, $CF_{com} = 1$, $ISC = 0$, $F = 1$, no cap: $r_{eff}(k) = k/N_{avg}$; $\mathcal{H}m_{f,a}^{'} = \max(r_{eff}(k) : \mathcal{TC}_{eff,k}^{adj} \geq r_{eff}(k)) =$ original hm; $\mathcal{H}m_a' = hm/\overline{hm}$.

□

6.4.6 Weighted average modified g index

Lemma 15 (Properties of $\mathcal{G}_a'$) The metric $\mathcal{G}_a'$ (Equations 15, 21, 22–23) is non-negative integer-bounded (by $\sqrt{\sum \mathcal{TC}^{eff}}$), field-normalized, emphasizes highly cited works via $\geq k^2$, and reduces to original g-index (normalized) under uniform conditions, single field.

Proof **Non-negativity:** $\mathcal{A}_{eff,f,a}(k) = \sum_{p'=1}^{k} \mathcal{TC}_{eff}^{adj} \geq 0$; $\max(k : \mathcal{A} \geq k^2) \geq 0$; normalization by $\overline{\mathcal{G}_f' > 0}$.
Normalization: Per-field g on effective cites; weighted average over fields.
Incorporation: Cumulative $\mathcal{A}_{eff}$ embeds factors; k^2 threshold favors influential papers.
Reduction: Uniform $\mathcal{W} = 1/N_p$, $CF_{com} = 1$, $ISC = 0$, $F = 1$, no cap: $\mathcal{A}_{eff}(k) = \sum(1/N_p)TC^{raw}$; $\mathcal{G}_{f,a}^{'} = g_f$ (original); $\mathcal{G}_a' = g_f/\overline{g_f}$. □

6.5 Proofs for overall Nm-Index formulation

We prove the validity of the Nm-index transformations (Equations 24–27) for a population of $N = 100,000$ scholars, ensuring comparable percentiles across skewed citations and tied indices.

Lemma 16 (Blom Transformation for Indices) For discrete, tied indices $\mathcal{H}f'_a, \mathcal{H}m'_a, \mathcal{G}'_a$, the transformation $P(x) = \frac{r(x)-0.375}{N+0.25}$ (fractional ranking) and $Nor_a(x) = \phi^{-1}(P(x))$ (Equation 25) yield $Per_a(x) = 100 \cdot P(x)$ (Equation 26) as uniform percentiles.

Proof • **Fractional ranking**: Mid-ranks $r(x) = \frac{\sum r_j}{k}$ for k tied values reduce bias [38].
• **Blom plotting**: $P(x) = \frac{r(x)-0.375}{N+0.25}$ estimates quantiles, **converging** to $U(0,1)$ by Glivenko–Cantelli theorem.
• **Normalization**: $\phi^{-1}(P(x)) \sim N(0,1)$ asymptotically (probability integral transform).
• **Percentile**: $Per_a(x) = 100 \cdot P(x) \sim U(0,100)$, **unbiased** for tied, skewed indices.
□

Lemma 17 (Hooked Power-Law Citation Transformation) For citation metrics $\mathcal{T}_a, \mathcal{S}_a, \mathcal{U}_a$ (hooked power-law, $\alpha > 2$), $\mathcal{L}(x) = \log_{10}(x+1)$ and $Nor_a(x) = \phi^{-1}(P(\mathcal{L}(x)))$ (Equation 25) yield $Per_a(x) = 100 \cdot \phi(Nor_a(x))$ (Equation 26) as uniform percentiles.

Proof • **Log transform**: $\mathcal{L}(x)$ reduces power-law skewness ($f(x) \propto x^{-\alpha}$) to Gumbel-like [36].
• **Quantile estimation**: $P(\mathcal{L}(x)) = \frac{r(\mathcal{L}(x))}{N+1} \sim U(0,1)$, **converging** with error $O_p(\sqrt{\log N/N})$ (Dvoretzky–Kiefer–Wolfowitz).
• **Normalization**: $\phi^{-1}(P(\mathcal{L}(x))) \sim N(0,1)$ (quantile function).
• **Percentile**: $Per_a(x) = 100 \cdot P(\mathcal{L}(x)) \sim U(0,100)$, **unbiased** across domains.
□

Lemma 18 (Log-Normal Citation Transformation) For log-normal citation metrics $\mathcal{T}_a, \mathcal{S}_a, \mathcal{U}_a$, $\mathcal{L}(x) = \log_{10}(x+1)$ and $Nor_a(x) = \frac{\mathcal{L}(x)-\overline{\mathcal{L}(x)}}{\sigma_{\mathcal{L}(x)}}$ (Equation 25) yield $Per_a(x) = 100 \cdot \phi(Nor_a(x))$ (Equation 26) as exact percentiles.

Proof • **Log transform**: For $x \sim \log \text{Normal}(\mu, \sigma^2)$, $\mathcal{L}(x) \approx \log_{10}(x) \sim N(\mu', \sigma'^2)$ [35].
• **Z-score**: $Nor_a(x) = \frac{\mathcal{L}(x)-\overline{\mathcal{L}(x)}}{\sigma_{\mathcal{L}(x)}} \sim N(0,1)$, **converging** by central limit theorem.
• **Percentile**: $Per_a(x) = 100 \cdot \phi(Nor_a(x))$ recovers exact quantile, **unbiased** for log-normal data.
□

Lemma 19 (Nm-Index Comparability) The Nm-index $Nm_a = \sum_{i=1}^{6} w_i Per_a(x_i)$, with $\sum w_i = 1$, $w_i = 1/6$ (Equation 27), is a comparable percentile score, robust to metric skewness.

Proof • **Uniform percentiles**: By Lemmas 16–18, $Per_a(x_i) \sim U(0,100)$.
• **Weighted average**: Nm_a is convex combination, **approximately uniform** ($E[Nm_a] = 50$, $Var \approx 100^2/12$ for independent x_i).
• **Comparability**: $Nm_a > Nm_b$ implies higher weighted rank, **invariant** to domain/career biases.
• **Robustness**: Equal $w_i = 1/6$ minimizes variance, **unbiased** for global ranking.
□

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